Monte Carlo determination of the thermal dimension $\Delta_\varepsilon(\Delta_\sigma)$ of the long-range Ising model in $d = 1, 2, 3$

Methods, data, results, limitations and reproduction

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Status of 15 October 2026

Abstract

We determine, by cluster Monte Carlo simulation, the scaling dimension Δ_ε of the leading $\mathbb{Z}_2$-even scalar of the ferromagnetic long-range Ising model with couplings $J(r) \propto r^{-(d+s)}$, as a function of the exactly known spin dimension $\Delta_\sigma = (d - s)/2$, in $d = 1, 2, 3$. Δ_ε is extracted from finite-size scaling of the scaled susceptibility and the Binder cumulants and their exact temperature derivatives. The analysis rules were reviewed by a separate AI system (OpenAI Codex with the model gpt-6-astra, which also drafted the simulation engines). In $d = 2, 3$ they were fixed before the production data were analyzed (one later amendment is noted below); in $d = 1$ they were revised after 8 000 of the 10 240 chains had been analyzed, and two gates were replaced after couplings had failed them (section 5). In $d = 1$ ($N \le 2^{19}$, 20 couplings) nine values pass the protocol, with statistical errors (2–5.5) $\times 10^{-4}$; they differ from the eight-loop resummed ε expansion by -0.0024 to $+0.0032$, up to 13 times their statistical errors. Their systematic error is the half-range of a family of fit models, a sensitivity scale rather than a standard deviation, and an audit by the same AI system finds the values at three couplings sensitive to the size cut or the correction model. At the three couplings closest to the Gaussian end ($\varepsilon \le 0.15$) the Monte Carlo values lie 0.0065–0.0104 below the expansion. In $d = 2$ ($L \le 1024$) the finite-size effective dimensions are precise ($\sigma \approx 0.002$) and agree with the resummed expansion to 0.0015 for $s = 1.2$–1.4, about as well as the expansion itself is determined there; no infinite-size value passes the protocol. In $d = 3$ ($L \le 128$) one value passes, $\Delta_\varepsilon(s{=}1.6) = 1.47314 \pm 0.00118\,(\text{stat}) \pm 0.00230\,(\text{sys})$, only under an amendment adopted after the data had been evaluated; it agrees with the expansion (1.4721). Near the short-range end the 3D extrapolations lie 0.022–0.033 above the expansion. We describe the model, algorithms, data, rules, results and limitations, and how to reproduce every number from the released records.

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1 Overview

1.1 Scope

This note is written for a physicist outside the project. It explains the problem and what is known exactly (section 2), the Monte Carlo and analysis methods in enough detail to re-implement them (sections 3–4), how and when the analysis rules were fixed (section 5), all results including negative ones (sections 6–9), the limitations and open problems (sections 10–11), and how to recompute every number from the released code and records (section 12). The work was carried out by AI agents under the direction of the author. The simulation engines (section 3) were drafted by GPT-6 astra (OpenAI, via ChatGPT) in discussions with the author and supplied as three separate packages: the initial 1D package (reference engine, campaign tools and analysis core), the specialized 1D engine, and the 2D/3D engine. The fit pipelines, protocols, evaluators and analyses were written by Claude (Anthropic) agents. Protocols and code were reviewed by a second AI system, OpenAI Codex with the model gpt-6-astra, the same model that drafted the engines, in sessions separate from the ones that wrote the protocols and fits. There was no human referee. Numbers in tables and figures are generated by script from the production records; nothing here is a new measurement except the diagnostics explicitly marked as such.

1.2 Headline results

Table 1: What the Monte Carlo establishes. "Passes the protocol" means the label ACCEPTED or ACCEPTED (σ_{sys}-DOMINATED) of section 4; it is a statement about the protocol (in $d = 3$ as amended by P1, section 7.2), not a guarantee of correctness. The theory comparison uses the eight-loop resummed ε expansion of the accompanying article [1] (CB8, section 6.5).

d	result	limitations
1	Nine of 20 couplings pass the protocol: Δ_ε with $\sigma_{\text{stat}} = (2.0\text{–}5.5) \times 10^{-4}$ and $\sigma_{\text{sys}} = (0.8\text{–}9.6) \times 10^{-3}$ (table 5); they differ from the resummed ε expansion by -0.0024 to $+0.0032$ (table 12).	σ_{sys} is a model-family half-range; the values at $\Delta_\sigma = 0.025$, 0.175, 0.225 are sensitive to the size cut or correction model; two labels (0.1625, 0.1727335) depend on the platform; quoted fits at 0.1 and 0.125 have $P_{\chi^2} \approx 5 \times 10^{-5}$; for $\Delta_\sigma \geq 0.0875$ the fit lies above the crossing extrapolation by a common $+0.0010(3)$ (section 6.4). At $\Delta_\sigma = 0.2125$–0.2375 the Monte Carlo values are 0.0065–0.0104 below the expansion.
2	Finite-size effective dimensions $\Delta_{\text{eff}}(U_4, 512 \to 1024) = 0.958$–$0.977$ with $\sigma \approx 0.002$ at all eight couplings; they agree with the resummed expansion to 0.0015 for $s = 1.2$–1.4, where the expansion itself depends on how its end point is imposed by up to 0.005 (table 18).	No infinite-size value passes the protocol. Two single-series extrapolations ($s = 1.05, 1.1$) lie 0.05–0.065 below the expansion; near both ends of the window the finite-size values drift away from it.
3	One value passes, $\Delta_\varepsilon(s{=}1.6) = 1.47314 \pm 0.00118 \pm 0.00230$, consistent with the expansion (1.4721) and with the finite-size $\Delta_{\text{eff}}(U_4, 64 \to 128) = 1.4715(31)$.	Accepted only under amendment P1, adopted after the data were evaluated, whose last code fixes are uninspected. Near the short-range end the extrapolations are 0.022–0.033 above the expansion, and the Y and U_4 estimates split.

2 The model and what is known exactly

2.1 Hamiltonian and conventions

Ising spins $S_i = \pm 1$ on a periodic hypercubic lattice of linear size L ($N = L^d$ sites) interact through the fully periodized power law

$$H = -\sum_{i<j} J_L(i-j)\, S_i S_j, \qquad J_L(\mathbf{r}) = \sum_{\mathbf{n}\in\mathbb{Z}^d} |\mathbf{r} + L\mathbf{n}|^{-d-s} \quad (\mathbf{r} \not\equiv 0 \bmod L), \tag{1}$$

with no Kac normalization (the nearest-neighbor coupling of the infinite lattice is one) and no extra short-range term. Inverse temperatures β, including critical ones, are quoted in this normalization. Below, a *coupling* always means one value of s (equivalently of Δ_σ) on the simulation grid. In $d = 1$ the lattice size is denoted N, and $J_N(r) = N^{-1-s}[\zeta(1+s, r/N) + \zeta(1+s, 1-r/N)]$ with the Hurwitz zeta function.

2.2 Long-range fixed points and their correction exponents

For $d/2 < s < s_*$ the critical point is the long-range fixed point [4, 5, 6]. Because the kinetic term is non-local, the spin field is not renormalized and

$$\Delta_\sigma = \frac{d-s}{2} \qquad \text{exactly,} \tag{2}$$

i.e. $\eta = 2 - s$. The thermal dimension Δ_ε (dimension of the leading $\mathbb{Z}_2$-even scalar, $1/\nu = d - \Delta_\varepsilon$) is the unknown. The ends of the window and the leading corrections to scaling are:

- *Gaussian end* $s \to d/2$: mean field, $\Delta_\varepsilon \to d - s = d/2$. With $\varepsilon = 2s - d$, $\Delta_\varepsilon = d - s + \varepsilon/3 + O(\varepsilon^2)$; the ϕ^4 coupling is nearly marginal and the leading correction exponent is $\omega = \varepsilon + O(\varepsilon^2)$, so corrections are slow ($\omega \approx \varepsilon$, which is 0.05 at the $d = 1$ coupling closest to this end and 0.1 in $d = 2, 3$).
- *Short-range end* $s \to s_* = 2 - \eta_{\mathrm{SR}}$ ($d = 2, 3$; Sak's criterion [5]): $s_* = 7/4$ in $d = 2$, $s_* \simeq 1.963702$ in $d = 3$, where $\Delta_\varepsilon \to \Delta_\varepsilon^{\mathrm{SR}} = 1$ ($d = 2$) and 1.412625 ($d = 3$; the 3D values from the conformal bootstrap [2]). Near s_* the fixed point is the short-range Ising fixed point weakly coupled to a generalized free field χ through $g \int \sigma\chi$ [7, 8]: with $\delta = (s_* - s)/2$, $dg/d\ell = \delta g - \beta_3 g^3$, where ℓ is the logarithm of the length scale, and $\gamma_\varepsilon \equiv \Delta_\varepsilon - \Delta_\varepsilon^{\mathrm{SR}} = c_\varepsilon g^2$, with $\beta_3 = 12.26(3)$ and $c_\varepsilon = 3.3(5)$ in $d = 3$ [8, eqs. (2.24), (3.41)] (and $\gamma_\varepsilon = O(\delta^2)$ in $d = 2$). The $\sigma\chi$ coupling approaches its fixed point as $\ell^{-(s_* - s)}$, a slow correction with exponent $s_* - s = 2\delta$ (0.05 at 2D $s = 1.7$, 0.014 at 3D $s = 1.95$). A lattice model with a pure power law starts at strong bare coupling, so the approach is a crossover rather than a small correction. It may contribute to the Y–U_4 splits seen near s_*, but in $d = 3$ the split grows with L, whereas at leading order this flow makes it shrink (sections 8, 9).
- $d = 1$, $s \to 1$ ($\Delta_\sigma \to 0$): the chain approaches the inverse-square point [9, 10], where the thermal perturbation becomes marginal ($\Delta_\varepsilon \to 1$); the leading correction exponent tends to zero together with the thermal exponent $y_t = 1 - \Delta_\varepsilon$ (at leading order of the Kosterlitz flow they are equal). Our fits near this end indeed give $\omega \approx y_t$ (table 7).

In $d = 2$ both ends have $\Delta_\varepsilon = 1$, so $\Delta_\varepsilon(s)$ dips below 1 in the interior; in $d = 3$ it interpolates between 1.5 and 1.4126.

2.3 Observables and the effective dimension

With $m = N^{-1} \sum_i S_i$ define the scaled second moment, the Binder cumulant [11] and higher moment ratios

$$Y = L^{2\Delta_\sigma} \langle m^2 \rangle, \quad U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}, \quad U_6 = \frac{\langle m^6 \rangle}{15\langle m^2 \rangle^3}, \quad U_8 = \frac{\langle m^8 \rangle}{105\langle m^2 \rangle^4}. \tag{3}$$

All are dimensionless at criticality up to corrections and are functions of the scaling variable $(\beta - \beta_c) L^{d - \Delta_\varepsilon}$. For a size-doubling pair $(L, 2L)$ let $\beta_\times$ be the crossing $U_4^{(L)}(\beta_\times) = U_4^{(2L)}(\beta_\times)$. At this same temperature

$$\Delta_{\mathrm{eff}}^{Q}(L) = d - \log_2 \frac{Q'_{2L}(\beta_\times)}{Q'_L(\beta_\times)}, \qquad Q \in \{U_4, U_6, U_8, Y\}, \quad Q' = \partial_\beta Q \tag{4}$$

(the quotient method [12]). Every Q gives Δ_ε as $L \to \infty$; their differences at finite L measure corrections to scaling. In $d = 1$ the Binder cumulant is written U and sizes are N.

3 Monte Carlo method

3.1 Cluster algorithm

All simulations use the extended-Poisson Swendsen–Wang [13] construction of Fukui and Todo [14], whose cost per sweep is $O(N)$ for long-range couplings (see also ref. [16]; for an application to the long-range Ising model, ref. [15]). Per sweep a Poisson number of candidate bond events is drawn with mean $\beta N \sum_{\mathbf{r}} J(\mathbf{r})$; each event picks a uniform source site and a displacement $\mathbf{r}$ with probability $\propto J(\mathbf{r})$, so each unordered pair receives events at rate $2\beta J_{ij}$. An event is kept only if the two spins are aligned; kept bonds are joined by union–find and every cluster receives an independent random sign. The number of kept events, K (counted with multiplicity, including repeated and cycle-closing events), makes the weight of a bond configuration proportional to β^K; cumulants of K therefore give exact β-derivatives and exact multi-temperature reweighting.

3.2 Engines

$d = 1$**, reference engine** (`code/1d/src/lri_mc.cpp`, C++17): Walker alias table over all $N - 1$ displacements, Mersenne twister, checkpoint/restart with atomic writes, one completion record per chain. Used for the pilot, the two-point campaign and stages 0, A and B (the $d = 1$ campaigns of table 4).

$d = 1$**, specialized engine** (`code/1d/src_1d/`): alias table for $r < 256$, geometric envelope with rejection for the tail, xoshiro256++ generator; 1.5–2.1× faster single core and 2.5–3.3× faster when 30 chains share a node. It agrees with the reference engine within statistics on 24 observable comparisons. Used for the N2 campaign (table 4).

$d = 2, 3$ (`code/2d3d_engine/code/src/`): exact periodic-image sampling without an L^d coupling table. Displacements in the core cube $[-16, 15]^d$ use exact weights via an alias table; the rest is a union of shells $C_{k+1} \setminus C_k$, $C_k = [-16 \cdot 2^k, 16 \cdot 2^k - 1]^d$, sampled from the envelope $(16 \cdot 2^k)^{-d-s}$ with rejection and folded modulo L; the 8 (2D) or 26 (3D) nearest offsets are separate Poisson processes. Records per sweep: K, improved estimators of m^2, m^4, m^6, m^8 from the cluster fractions c_j ($S_{2k} = \sum_j c_j^{2k}$: $m_2 = S_2$, $m_4 = 3S_2^2 - 2S_4$, $m_6 = 15S_2^3 - 30S_2S_4 + 16S_6$, $m_8 = 105S_2^4 - 420S_2^2S_4 + 140S_4^2 + 448S_2S_6 - 272S_8$), the sampled m^2, the largest cluster and the number of clusters. The engine was checked against exact enumeration of 4×4 and 2^3 systems and against the reference engine, and restarts reproduce bitwise.

3.3 Derivatives and reweighting

For a graph observable A, $\partial_\beta \langle A \rangle = \mathrm{Cov}(A, K)/\beta$ and $\partial_\beta^2 \langle A \rangle = [\langle A(K - \bar{K})^2 \rangle - \langle A \rangle \mathrm{Var} K - \mathrm{Cov}(A, K)]/\beta^2$ (the last term arises because the weight is β^K). In $d = 1$ the analysis keeps, per (N, β), the six quantities (Y, Y', Y'', U, U', U'') with a 6×6 jackknife covariance from blocks of 2048, 4096 (primary) and 8192 sweeps. In $d = 2, 3$ each chain is packed into histograms in K at blocks of 1000, 2000 (primary) and 4000 sweeps, and the five simulated temperatures of a size are combined by multi-temperature reweighting (MBAR [17], the multiple-histogram method [18]) with the exact factor $(\beta/\beta_0)^K$.

3.4 Crossings and bootstrap in $d = 2, 3$

For each pre-registered pair, the crossing $\beta_\times$ is the unique root of $U_4^{(2L)} - U_4^{(L)}$ inside the shared temperature bracket (exactly one root required, reweighting effective sample size ≥ 0.2). Errors come from 500 joint bootstrap draws (seed 571239): in each draw the blocks of every chain are resampled once (multinomial weights), the same resample serves both pairs that contain that size, and crossings and slopes are recomputed; the draws give the full covariance across pairs and observables.

3.5 Equilibration and provenance

Every (L, β) group has hot and cold replicas with 5×10^4 warm-up sweeps. In $d = 2, 3$ the equilibration rule of a coupling requires (i) every chain's block mean of m^2 within $|z| \leq 4$ of its group, and at most 2 % of the coupling's chains beyond $|z| = 3$; and (ii) no group in which half or more of the chains fail the drift test (first against second half of the chain, $|z| \leq 4$), the condition block length $\geq 20\,\tau_{\mathrm{int}}$ at every block size used, or finiteness. In $d = 3$ the rule is also applied to the reblocking records. In $d = 1$ (protocol v3) every group must have its hot–cold difference and every replica's first–second-half difference within $|z| \leq 4$ and τ_{int} below block/20; across the coupling at most 2 % of the groups may exceed $|z| = 3$ in the hot–cold test and at most 2 % of the replicas in the half-chain test. A failure labels the coupling EQUILIBRATION FAILED; no chain is removed. In the preflight runs (short runs at the production sizes, before each production, that check rates, autocorrelation times and crossing brackets) τ_{int} was at most 17 sweeps ($d = 1$), 29 ($d = 2$, $L = 1024$) and 34 ($d = 3$, $L = 128$). Each chain writes a completion record; snapshot manifests list the SHA-256 of every summary, block archive, registry and code file; output directories are created afresh and never overwritten.

4 How a label is assigned

Each coupling receives exactly one label from a fixed decision table, applied in the order of table 2; the first condition that applies gives the label. Only the two ACCEPTED labels denote values that passed the protocol (in $d = 2, 3$ the frozen rule or, where it is calibrated, amendment P1; section 7.2). A *gate* is a pass/fail condition on a fit or an extrapolation series; a $d = 1$ fit that passes its gates is *quotable* (section 6.2), and a $d = 2, 3$ series that passes its gates is *eligible* (section 7).

The two errors. σ_{stat} is a bootstrap error of the quoted value. σ_{sys} is half the range of the quoted values over a family of admissible fit models (in $d = 1$) or extrapolation series (in $d = 2, 3$): a *model-family sensitivity scale*, not a standard deviation and not a bound. It covers only the variation inside that family, it is symmetric about the quoted value although the family's excursions need not be, and there is no justified rule for adding it to σ_{stat} in quadrature. Values are written $v(\sigma_{\mathrm{stat}})(\sigma_{\mathrm{sys}})$ in units of the last digits.

Table 2: The decision labels and what triggers them. "Cross-check": two extrapolations of the same data (in $d = 1$ the global fit and the crossing series, in $d = 2, 3$ the U_4 and Y series) must agree within $2\sigma_D$, σ_D being the error of their paired difference.

label	trigger in $d = 1$	trigger in $d = 2, 3$
EQUILIBRATION FAILED	a group fails the equilibration rule	same
COVERAGE-LIMITED	a pre-registered size or temperature missing	a pre-registered pair missing
MODEL-LIMITED	no fit passes the quality gates (finite-size values reported)	no extrapolation series is eligible
CONDITIONAL	a quoted fit exists but σ_{sys} is not available: fewer than four quotable fits, or fewer than two minimum sizes or two variants among them	exactly one eligible series, so no σ_{sys}
REBLOCKING FAILED/UNAVAILABLE	σ_{stat} at blocks 2048 or 8192 outside $\pm 30\,\%$ of the primary	same at blocks 1000 or 4000
CROSS-CHECK UNAVAILABLE	fewer than two eligible crossing series	U_4 and Y not both eligible
CROSS-CHECK FAILED	global fit and median crossing extrapolation differ by more than $2\sigma_D$	extrapolated U_4 and Y differ by more than $2\sigma_D$
ACCEPTED	none of the above	none of the above
ACCEPTED (σ_{sys}-DOMINATED)	accepted and $\sigma_{\mathrm{sys}} > 3\times$ the median σ_{stat} of the quotable fits	–

Suffix "[σ]" ($d = 2, 3$): $\sigma(\Delta_{\mathrm{eff}}^{U_4})$ of the largest pair exceeds the target 0.003. A label is prefixed NON-PRODUCTION when the provenance checks of its inputs do not pass. Tables abbreviate cross-check to x-check, unavailable to unavail., dominated to dom., reblocking to reblock. and equilibration to equil.

5 How the rules were fixed

The aim was to write each analysis protocol, and have it reviewed, before the data it governs were analyzed, with selection rules blind to Δ_ε. Table 3 records how far this held. Reviews were done by the second AI system named in section 1: plan reviews of the protocol texts, each ending with the verdict "approved" (no unresolved material defect in the reviewed revision) or "revise", and inspections of the code that implements them. The protocol texts, their hashes and their review status are released with the code (appendix D); `REVIEWS.md` summarizes every review round and verdict.

$d = 1$: rules changed after failures. Two gates of v2 were replaced in v3 after couplings had failed them on these data. The absolute cap on σ_{sys} (0.0015 for $\Delta_\sigma \le 1/16$) failed at all four couplings $\Delta_\sigma \le 0.05$; v3 has no absolute cap and labels a large σ_{sys} instead (ACCEPTED (σ_{sys}-DOMINATED)). The per-group equilibration rule failed at 0.0875 and 0.125, each on one group of 50 ($|z| = 3.04$ and 3.05); v3 applies a population rule (no group above 4σ, at most $2\,\%$ of groups above 3σ). Both changes have a statistical reason (a cap fixed on two couplings; a per-group rule that a 3σ excursion fails at about $13\,\%$ of couplings by chance), but four of the nine ACCEPTED labels depend on them: 0.025 and 0.05 on the cap, 0.0875 and 0.125 on the equilibration rule.

$d = 1$: the N0 diagnostic. N0 tested, at $\Delta_\sigma = 0.175$, whether the planned N2 window could be described by the fit polynomials, and so decided whether N2 ran and with which window. It was evaluated under three rules, the second and third written after its data had been seen. The pre-registered rule gave INCONCLUSIVE; its prescribed remedy, a retest at half the temperature spacing, was not run. The second rule gave FAIL, which would have stopped N2 and returned the plan to review; that failure came from selecting too low a polynomial degree. The third rule, a joint polynomial test of values and derivatives, gave PASS and released N2 with the reduced window $|x| \le 0.35$.

6 $d = 1$

Table 3: When the rules were fixed, relative to the data they govern.

rules	review status	data seen when the rules were fixed
$d = 1$ (FIT_MODEL_V3)	approved at revision 14 (2026-09-19). The first evaluation of all 20 couplings with v3 (called N1, 2026-09-18) ran under revision 8, which had not been reviewed (the review of revision 7 had asked for changes). The last fixes to the implementation are uninspected.	The previous protocol (v2, approved 2026-09-16) had been applied to stages A and B (8 000 of the 10 240 chains of the final fits). v3 was written after those results; its temperature windows (the frozen $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$) come from fits of those data, and two v2 gates were replaced after couplings had failed them (below). The rule of the N0 diagnostic (table 4) was amended twice after its data had been seen (below). Only the N2 data (2 240 chains) were taken after approval.
$d = 2$ (FIT_MODEL_2D)	reviewed, never approved; the last fixes applied without a further round	Frozen 2026-09-20, before the production; stage-0 and preflight data had been seen.
$d = 3$ (FIT_MODEL_3D)	revision 5 approved; production under revision 7, whose review returned one finding, fixed without a further round	Revision 7 changed one preflight condition (a τ_{int} cap of 30 to the protocol's own block/$\tau \geq 20$ rule) after the preflight had run; its record also showed preliminary Δ_{eff} values. No production chain had run.
$d = 2, 3$ amendment P1	approved; code inspected twice, the second round's fixes uninspected	Drafted, reviewed and approved after both production evaluations were known, with its effect on every label already computed (section 7.2).

6.1 Grid and campaigns

Twenty couplings $\Delta_\sigma = k/80$ ($k = 1 \dots 19$) and $\Delta_\sigma = 0.1727335$ ($s = 0.654533$, the coupling studied in ref. [20]). Every chain of the final fits (stages A, B and N2) has 5×10^4 warm-up and 2×10^6 measured sweeps, every (N, β) group has 8 hot/cold replicas, and each size has five temperatures; the other campaigns of table 4 differ.

Table 4: $d = 1$ campaigns. Only stages A, B and N2 enter the final fits: 10 240 chains, 2.05×10^{10} measured sweeps, 0.655 TB of raw trajectories.

campaign	design	chains	in final fits?
pilot	bracketing at $\Delta_\sigma = 0.1, 1/16$, $N \leq 32\,768$	–	no
two-point	$\Delta_\sigma = 0.1, 1/16$, $N = 2^{10} \dots 2^{19}$, half-width $h(N) = 0.005(N/1024)^{-0.35}$	800	no
stage 0	bracket finder for the temperature centers	–	no
stage A	20 points, $N = 2^{10} \dots 2^{16}$	5 600	yes
stage B	20 points, $N = 2^{17} \dots 2^{19}$	2 400	yes
N0	diagnostic at $\Delta_\sigma = 0.175$, $N = 2^{16}, 2^{18}$, 9 temperatures	72	no
N2	14 points ($\Delta_\sigma = 1/16$ and ≥ 0.1), $N = 2^{16} \dots 2^{19}$, temperatures matched in the scaling variable, specialized engine	2 240	yes

6.2 Fit model and selection

Model. For $Q \in \{Y, U\}$, $r = N/1024$ and the scaling variable $x = 100\,\tau(\beta)\, r^{1-\Delta_\varepsilon}$, where $\tau = (\beta - \beta_c)[1 + a_1(\beta - \beta_c)]$ for field order $k_\tau = 2$ ($a_1 = 0$ for $k_\tau = 1$) and the factor 100 only makes x of order one,

$$Q_N(\beta) = F_Q(x) + r^{-\omega} G_Q(x)\ [+\, r^{-2\omega} H_Q(x)]\ [+\, r^{-s}(b_0 + b_1(\beta - \beta_c)) \text{ on } Y]\ [\times(1 + c_Q/\ln(N/256)) \text{ on } F_Q]. \quad (5)$$

F_Q is a polynomial of degree $p = 3 \dots 7$ and G_Q, H_Q are quadratic. The four *variants* are: `leading` (no bracketed term), `second` (the $r^{-2\omega}$ term), `background` (the analytic r^{-s} background of Y) and `log` (the logarithmic factor; only for $\Delta_\sigma \leq 0.05$). Each (N, β) contributes six rows to the fit: Y, Y', Y'', U, U', U'', the derivatives following from the model by the chain rule, with the full 6×6 covariance. The nonlinear parameters $(\beta_c, \Delta_\varepsilon, \omega, a_1[, c_Q])$ are fitted by profiling over all linear amplitudes.

Configurations. Data are selected by windows $|x| \le x_{\max} \in \{0.35, 0.45, 0.6\}$, computed with frozen $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$ (appendix A), and minimum sizes $N_{\min} \in \{4096, 8192, 16384\}$; at least four sizes with three temperatures each must remain. Windows × variants × $N_{\min} \times k_\tau$ give 54 (72 with `log`) configurations per coupling; at $\Delta_\sigma = 0.075$ and 0.0875 the window 0.35 leaves too few temperatures, so these couplings have 36, and the 20 couplings have 1 116 in all.

Degree ladder. For each configuration the degree starts at $p = 3$ and is raised while an F-test prefers $p+1$ (F-test probability below 0.01); the ladder stops at the first missing or flagged fit (a fit is flagged if a nonlinear parameter sits on its search bound, a matrix is rank deficient or the optimizer fails). A fit at $p+1$ must agree with p to $3\sigma_{\mathrm{stat}}$. We write P_{χ^2} for the χ^2 probability of a fit and keep p for the degree.

Quotable fits. A configuration's fit is quotable if $\chi^2/\mathrm{dof} \le 1.5$, it is unflagged, the mean |pull| of each observable is ≤ 1.2, at least 270 of 300 bootstrap refits are valid, and a test of residual structure passes (runs of pull signs and lag-one correlation of pulls along β, with thresholds calibrated on synthetic data drawn from the fit itself).

The quoted value is the quotable fit with, in order of preference, the smallest $N_{\min}$, the fewest parameters, the window 0.45 before 0.6 before 0.35, the variant `leading` before `background` before `second` before `log`, the lower k_τ, the lower degree. σ_{stat} is half the 16–84 % range of its 300 bootstrap refits, in which the blocks of every chain are resampled and all nonlinear parameters are refitted. σ_{sys} is half the range over all quotable fits and requires at least four fits spanning two $N_{\min}$ and two variants.

Crossing cross-check. Independently of the global fit, local polynomials in β through the in-window temperatures of each size give the U-crossings of the four largest usable doubling pairs (a size is usable if its local fit is unflagged with $\chi^2/\mathrm{dof} \le 2$, a pair if both sizes are usable and the crossing lies inside their common bracket). There are up to four *crossing series*, one per crossing window $|x| \le x_c \in \{0.45, 0.6\}$ and local-degree rule (the degree p^* chosen by an F-test, or p^*+1), named `xc0.45_star`, `xc0.45_plus`, `xc0.6_star` and `xc0.6_plus`. Each is extrapolated as $\Delta_{\mathrm{eff}}^{Q}(N) = \Delta_\varepsilon^\infty + c_Q N^{-\omega'}$ with $\omega' \in [0.2, 2]$ profiled on a grid of step 0.01 in every bootstrap draw; a series is eligible if $\chi^2/\mathrm{dof} \le 2$ and ω' is off the grid bounds. With at least two eligible series, the test is $|\Delta_\varepsilon^{\mathrm{fit}} - \mathrm{median}_E\, \Delta_\varepsilon^\infty| \le 2\sigma_D$, where, as pre-registered in the protocol, σ_D is the half 16–84 % width of the paired bootstrap difference *enlarged in quadrature* by half the spread of the central values of the eligible series. This makes σ_D itself a mixed statistical and sensitivity quantity; table 7 gives z with and without the spread term. The crossing value is quoted as $\Delta_\varepsilon^\infty(\sigma_\infty)(\Delta_{\omega'})$: σ_∞ is the bootstrap error of $\Delta_\varepsilon^\infty$ with ω' re-profiled in every draw, and $\Delta_{\omega'}$ is a spread of $\Delta_\varepsilon^\infty$ (not of ω'): the between-series half-spread plus the half-range of $\Delta_\varepsilon^\infty$ over the union of the series' $\Delta\chi^2 \le 1$ intervals of ω'. It counts the between-series spread twice and is an enlarged sensitivity measure.

Worked examples. At $\Delta_\sigma = 0.1727335$, 4 of 54 configurations are quotable; the preferred one (window 0.35, variant `second`, $N_{\min} = 4096$, $k_\tau = 2$, $p = 5$) gives 0.48368(52); the four quotable values span $\pm 0.00078 = \sigma_{\mathrm{sys}}$; four crossing series are eligible, their median lies 0.00013 below the fit with $\sigma_D = 0.00078$, so the cross-check passes and the label is ACCEPTED. At $\Delta_\sigma = 0.15$, 11 configurations are quotable, the quoted value is 0.49325(41) with $\sigma_{\mathrm{sys}} = 0.00306$, and the crossing median lies 0.00143 below with $\sigma_D = 0.00068$: CROSS-CHECK FAILED.

6.3 Results

Reading the results.

Near $s \to 1$ ($\Delta_\sigma \le 0.05$). 68–72 configurations are quotable, but $\sigma_{\mathrm{sys}} = 4$–$20 \times \sigma_{\mathrm{stat}}$: the fitted correction exponents ($\omega \approx 0.26$–0.40) and the variants do not settle the correction sector, as expected near a marginal point.

Middle ($0.0625 \le \Delta_\sigma \le 0.1375$). Six of seven couplings pass, with $\sigma_{\mathrm{stat}} = (2\text{–}4) \times 10^{-4}$ and $\sigma_{\mathrm{sys}} = (1.4\text{–}2.4) \times 10^{-3}$. At $\Delta_\sigma = 1/16$ the value is stable across campaigns: 0.59003 (final fit, with N2), 0.59046 (stages A and B alone, under protocol v2) and 0.59071 (the separate two-point campaign); but the crossing series fail their own goodness-of-fit gate, hence CROSS-CHECK UNAVAILABLE.

Towards the Gaussian end ($\Delta_\sigma \ge 0.15$). $\Delta_\sigma = 0.1727335$ passes. At 0.15 and 0.175 the cross-check fails by 2.1 and $2.0\sigma_D$; at 0.1625, 0.1875 and 0.225 only fits of variant `second` with $k_\tau = 2$ are quotable (CONDITIONAL); at 0.2, 0.2125 and 0.2375 no fit is quotable, because the correction exponent $\omega \approx \varepsilon \le 0.2$

Table 5: $d = 1$ results. Δ_ε is value(σ_{stat})(σ_{sys}) in units of 10^{-5}; †: σ_{sys} not available. "quot.": number of quotable fits; quoted configuration: window, variant, $N_{\min}$, field order and polynomial degree (e.g. `x0.45_leading_Nmin4096_K1 d4`: $|x| \leq 0.45$, variant `leading`, $N_{\min} = 4096$, $k_\tau = 1$, $p = 4$); crossing value $\Delta_\varepsilon^\infty(\sigma_\infty)(\Delta_{\omega'})$ (section 6.2). Published values: UGA [19], APR [20].

Δ_σ	s	decision	Δ_ε (stat)(sys)	quot.	quoted configuration	crossing value	published MC
0.0125	0.975	CROSS-CHECK UNAVAILABLE	0.78404(77)(1003)	72	x0.45_leading_Nmin4096_K1 d3	–	
0.025	0.95	ACCEPTED (σ_{sys}-DOMINATED)	0.71115(49)(959)	68	x0.45_leading_Nmin4096_K1 d4	0.70906(902)(1509)	
0.0375	0.925	CROSS-CHECK UNAVAILABLE	0.65904(51)(425)	71	x0.45_leading_Nmin4096_K1 d4	–	
0.05	0.9	ACCEPTED	0.61987(55)(220)	72	x0.45_leading_Nmin4096_K1 d4	0.62393(348)(773)	0.60(1) UGA
0.0625	0.875	CROSS-CHECK UNAVAILABLE	0.59003(35)(150)	50	x0.45_leading_Nmin4096_K1 d5	–	0.5876(13) APR
0.075	0.85	ACCEPTED	0.56694(37)(139)	30	x0.45_leading_Nmin4096_K2 d4	0.56843(207)(1044)	
0.0875	0.825	ACCEPTED	0.54761(32)(194)	34	x0.45_leading_Nmin4096_K1 d5	0.54711(113)(328)	
0.1	0.8	ACCEPTED (σ_{sys}-DOMINATED)	0.53248(20)(242)	48	x0.45_leading_Nmin4096_K1 d5	0.52975(135)(248)	0.53(2) UGA
0.1125	0.775	ACCEPTED (σ_{sys}-DOMINATED)	0.52020(25)(217)	41	x0.45_leading_Nmin4096_K2 d5	0.51790(86)(944)	
0.125	0.75	ACCEPTED	0.50710(35)(193)	9	x0.35_background_Nmin4096_K2 d5	0.50640(172)(392)	
0.1375	0.725	ACCEPTED (σ_{sys}-DOMINATED)	0.49856(30)(180)	17	x0.45_background_Nmin4096_K2 d4	0.49783(94)(85)	
0.15	0.7	CROSS-CHECK FAILED	0.49325(41)(306)	11	x0.35_background_Nmin4096_K2 d5	0.49183(71)(70)	0.49(2) UGA
0.1625	0.675	CONDITIONAL	0.48761(58)(†)	5	x0.35_second_Nmin4096_K2 d5	0.48732(75)(297)	
0.1727335	0.6545	ACCEPTED	0.48368(52)(78)	4	x0.35_second_Nmin4096_K2 d5	0.48355(62)(84)	0.494(14) APR
0.175	0.65	CROSS-CHECK FAILED	0.48460(35)(67)	6	x0.45_second_Nmin4096_K2 d6	0.48304(81)(1)	
0.1875	0.625	CONDITIONAL	0.48263(30)(†)	4	x0.45_second_Nmin4096_K2 d5	–	
0.2	0.6	MODEL-LIMITED	–	0	–	–	0.50(2) UGA
0.2125	0.575	MODEL-LIMITED	–	0	–	–	
0.225	0.55	CONDITIONAL	0.47912(68)(†)	1	x0.35_second_Nmin16384_K2 d7	–	
0.2375	0.525	MODEL-LIMITED	–	0	–	–	

Table 6: $d = 1$: effective dimensions $\Delta_{\mathrm{eff}}(U)/\Delta_{\mathrm{eff}}(Y)$ at the U-crossing of the two largest usable doubling pairs (errors in units of 10^{-4}, 300 draws). For the MODEL-LIMITED couplings these finite-size values are the only result.

Δ_σ	pair 1	pair 2
0.0125	128k–256k: 0.7606(57) / 0.7583(26)	256k–512k: 0.7747(61) / 0.7671(28)
0.025	128k–256k: 0.7034(32) / 0.7015(17)	256k–512k: 0.7094(37) / 0.7044(20)
0.0375	128k–256k: 0.6617(31) / 0.6551(19)	256k–512k: 0.6545(34) / 0.6528(20)
0.05	128k–256k: 0.6211(26) / 0.6174(17)	256k–512k: 0.6185(26) / 0.6173(17)
0.0625	128k–256k: 0.5889(13) / 0.5850(9)	256k–512k: 0.5921(14) / 0.5893(10)
0.075	128k–256k: 0.5670(18) / 0.5628(14)	256k–512k: 0.5676(20) / 0.5659(16)
0.0875	128k–256k: 0.5452(17) / 0.5418(13)	256k–512k: 0.5472(19) / 0.5440(15)
0.1	128k–256k: 0.5302(12) / 0.5251(10)	256k–512k: 0.5296(13) / 0.5279(11)
0.1125	128k–256k: 0.5194(12) / 0.5144(10)	256k–512k: 0.5185(13) / 0.5140(11)
0.125	128k–256k: 0.5058(12) / 0.5003(10)	256k–512k: 0.5084(13) / 0.5020(12)
0.1375	128k–256k: 0.5005(12) / 0.4921(11)	256k–512k: 0.4989(13) / 0.4918(12)
0.15	128k–256k: 0.4942(13) / 0.4819(12)	256k–512k: 0.4930(14) / 0.4837(13)
0.1625	128k–256k: 0.4875(12) / 0.4746(11)	256k–512k: 0.4889(14) / 0.4771(13)
0.1727335	128k–256k: 0.4835(13) / 0.4673(14)	256k–512k: 0.4852(14) / 0.4715(13)
0.175	128k–256k: 0.4839(14) / 0.4668(13)	256k–512k: 0.4851(14) / 0.4700(13)
0.1875	128k–256k: 0.4846(14) / 0.4645(13)	256k–512k: 0.4803(14) / 0.4625(14)
0.2	128k–256k: 0.4866(14) / 0.4587(13)	256k–512k: 0.4819(15) / 0.4570(15)
0.2125	128k–256k: 0.4855(15) / 0.4499(15)	256k–512k: 0.4784(15) / 0.4518(15)
0.225	128k–256k: 0.4814(15) / 0.4405(16)	256k–512k: 0.4844(16) / 0.4464(16)
0.2375	16k–32k: 0.4825(23) / 0.4150(19)	128k–256k: 0.4826(15) / 0.4307(16)

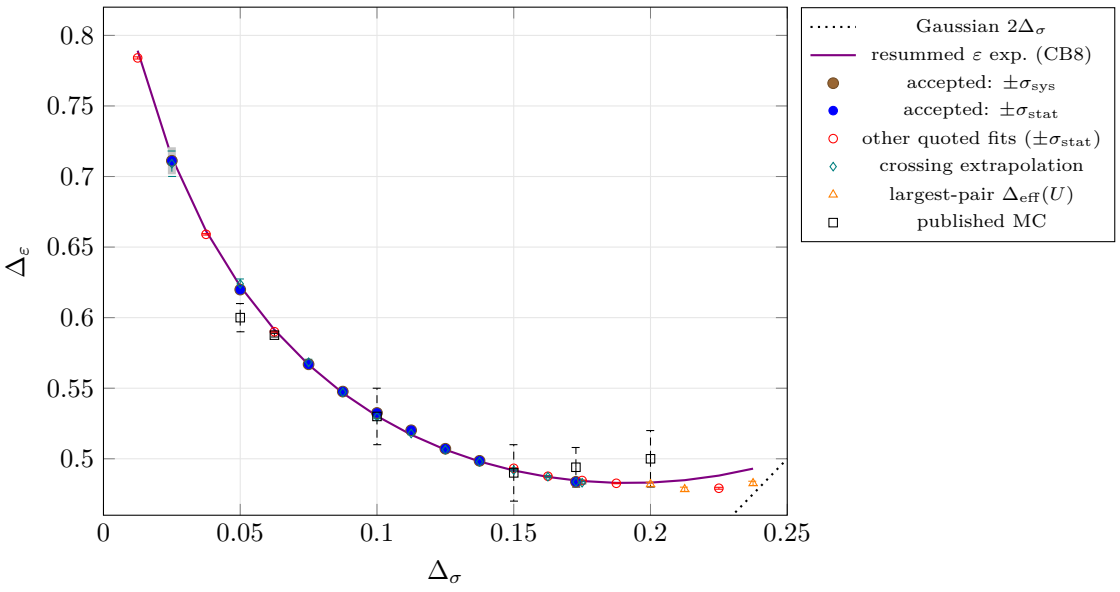


Figure 1: $d = 1$: $\Delta_\varepsilon(\Delta_\sigma)$. Values that passed the protocol with σ_{stat} (blue bars) and, separately, the model-family half-range σ_{sys} (gray bands), which is not a standard deviation; other quoted fits; crossing extrapolations with σ_∞; largest-pair effective dimensions where no fit is quotable; published Monte Carlo values; and the resummed ε expansion. Table 12 gives the differences.

is too small to separate from the scaling function over $N \leq 2^{19}$. There the finite-size values of table 6 are reported; they are not infinite-size estimates.

6.4 Robustness of the $d = 1$ results

Table 7 collects, per coupling, the quantities that bear on how much a value or label can be trusted; the paragraphs below explain them.

Table 7: $d = 1$ robustness. n_q: quotable fits; P_{χ^2}: χ^2 probability of the quoted fit; ω: its correction exponent; fit − crossing: quoted value minus the median crossing extrapolation; z: that difference over σ_D as pre-registered (with the spread term) and over the bootstrap part alone; P: label platform-sensitive (section 6.4.1).

Δ_σ	decision	n_q	P_{χ^2}	ω	fit − crossing	z	z (bootstrap)	
0.0125	X-CHECK UNAVAIL.	72	0.01	0.262	–	–	–	
0.025	ACCEPTED (σ_{sys}-DOM.)	68	3.1×10^{-4}	0.309	+0.00209	+0.23	+0.24	
0.0375	X-CHECK UNAVAIL.	71	0.21	0.348	–	–	–	
0.05	ACCEPTED	72	0.36	0.397	-0.00406	-0.92	-1.16	
0.0625	X-CHECK UNAVAIL.	50	0.29	0.427	–	–	–	
0.075	ACCEPTED	30	0.51	0.418	-0.00148	-0.33	-0.70	
0.0875	ACCEPTED	34	0.41	0.385	+0.00050	+0.35	+0.50	
0.1	ACCEPTED (σ_{sys}-DOM.)	48	5.2×10^{-5}	0.367	+0.00273	+1.73	+2.13	
0.1125	ACCEPTED (σ_{sys}-DOM.)	41	1.5×10^{-3}	0.368	+0.00230	+0.61	+2.73	
0.125	ACCEPTED	9	4.6×10^{-5}	0.368	+0.00070	+0.31	+0.42	
0.1375	ACCEPTED (σ_{sys}-DOM.)	17	0.03	0.365	+0.00073	+0.77	+0.79	
0.15	X-CHECK FAILED	11	2.1×10^{-3}	0.320	+0.00143	+2.10	+2.24	
0.1625	CONDITIONAL	5	0.05	0.194	+0.00029	+0.22	+0.38	P
0.1727335	ACCEPTED	4	0.08	0.195	+0.00013	+0.17	+0.20	P
0.175	X-CHECK FAILED	6	0.03	0.192	+0.00155	+2.00	+2.00	
0.1875	CONDITIONAL	4	1.5×10^{-4}	0.155	–	–	–	
0.2	MODEL-LIMITED	0	–	–	–	–	–	
0.2125	MODEL-LIMITED	0	–	–	–	–	–	
0.225	CONDITIONAL	1	7.4×10^{-4}	0.083	–	–	–	
0.2375	MODEL-LIMITED	0	–	–	–	–	–	

Goodness of fit. The quality gate $\chi^2/\text{dof} \leq 1.5$ admits, at about 200 degrees of freedom, fits with $P_{\chi^2} \sim 10^{-5}$. The quoted fits at $\Delta_\sigma = 0.1$ and 0.125 have $P_{\chi^2} \approx 5 \times 10^{-5}$, at 0.025 $P_{\chi^2} = 3 \times 10^{-4}$, and at 0.1125 $P_{\chi^2} = 1.5 \times 10^{-3}$. The preference for the smallest $N_{\min}$ selects the member of the family with the most data, which is also often the one with the largest χ^2. For these couplings σ_{stat} understates the point-to-point scatter, and the values should be read with σ_{sys}.

The cross-check and its σ_D. Without the spread term in σ_D the test would fail at $\Delta_\sigma = 0.1$ ($z = 2.13$) and 0.1125 ($z = 2.73$), both labeled ACCEPTED. Moreover, for all nine couplings with $\Delta_\sigma \geq 0.0875$ and a cross-check the fit lies above the crossing extrapolation: the inverse-variance mean difference is $+0.00105 \pm 0.00028$ with $\chi^2 = 8.1$ for 8 degrees of freedom about it. (Inverse-variance weights from the bootstrap part.) The two failures at 0.15 and 0.175 are among the points with the smallest σ_D, not isolated anomalies. The global fit and the crossings use different correction exponents (for example $\omega = 0.32$ and $\omega' = 0.31$–0.32 at 0.15, $\omega = 0.19$ and $\omega' = 0.27$ at 0.175) and the same data, so their comparison is a consistency check of two extrapolation models, not an independent measurement.

The failures and the ω' grid. The failures at 0.15 and 0.175 are not an artifact of conditional errors: the crossing bootstrap refits ω' in every draw, and fixing it at its central value would make σ_D smaller (table 8). They are, however, numerically fragile: refining only the ω' grid from step 0.01 to 0.001 and 0.0001, with everything else fixed, moves the ratio $|\Delta_\varepsilon^{\text{fit}} - \Delta_\varepsilon^{\infty}|/\sigma_D$ from 2.101 to 1.884 and 1.907 at 0.15, and from 2.003 to 1.897 and 1.931 at 0.175, below the threshold. The labels stand under the frozen protocol (step 0.01); whether a fit–crossing difference of this size is real is unresolved, and the common offset above suggests a systematic difference between the two extrapolation models.

Table 8: The cross-check at $\Delta_\sigma = 0.15, 0.175$ with ω' refitted per draw (as recorded) and fixed at its central value. Last column: central ω' (16–84 % of the per-draw values) of each eligible crossing series.

Δ_σ	fit − crossing	σ_D (per draw)	z	σ_D (fixed ω')	z	series
0.15	+0.00143	0.00068	+2.10	0.00062	+2.30	xc0.45_star: 0.32 (0.31–0.32), xc0.45_plus: 0.31 (0.30–0.32)
0.175	+0.00155	0.00078	+2.00	0.00067	+2.30	xc0.6_star: 0.27 (0.26–0.27), xc0.6_plus: 0.27 (0.26–0.27)

Audit. On 2026-09-22 the second AI system audited the finished analysis, with diagnostic refits (no new data) on the same records and bootstrap draws. It found no defect in the sampler, estimators or implementation, and these sensitivities:

- The quoted values at $\Delta_\sigma = 0.1$, 0.1375 and 0.1727335 are stable under stronger size cuts ($N \geq 32\,768$, $N \geq 65\,536$); at 0.1727335 the cuts give 0.48421 and 0.48386 against 0.48368(52).
- At 0.025 a model outside the family (analytic background plus second power together) gives 0.69348, below the family's lower end 0.69686, a shift of 7.0 paired-bootstrap scales (half the 16–84 % width of the matched difference). At 0.175, $N \geq 32\,768$ shifts the value by -0.00237 (2.2 scales); at 0.225, $N \geq 65\,536$ shifts it by $+0.00483$ (3.7 scales).
- The family is asymmetric: at 0.1375 it extends 0.00020 below and 0.00340 above the quoted value.
- A regular magnetic background gives an N^{-s} correction to U as well as to Y, but the `background` variant puts it only on Y. In the six cases the audit reports (0.025, 0.1, 0.1375, 0.1727335, 0.175, 0.225; variant with background and second power), adding it moves the central value by at most 0.0021 (at 0.025). At 0.05 such a model gives 0.61602, 0.0039 below the quoted value and below the family range [0.61845, 0.62286] (central fit only, not bootstrapped).

The degree ladder. In 160 of the 1 116 configurations, at 15 of the 20 couplings, the ladder cannot take its first step ($p = 3$ against $p = 4$) because one of the two fits is missing or flagged. In 140 of them it is the $p = 3$ fit, although that fit is typically a gross misfit that the F-test would reject by many orders of magnitude; in the other 20 it is the $p = 4$ fit. n_q and σ_{sys} are computed on the families trimmed in this way. At two couplings this decides the label (section 6.4.1). A sensitivity study with a ladder that does not stop on a flagged lower-degree fit is reported in section 6.4.2.

6.4.1 Reproducibility of the fitting stage across platforms

The fit inputs of every coupling (merged summaries and block archives) are released, so the whole fitting stage can be rerun (`make refit1d`). It was rerun for all 20 couplings on 7 October 2026 on a workstation (Intel i9-13900K; Python 3.12.3, NumPy 1.26.4, SciPy 1.11.4, OpenBLAS) with the production arguments and compared with the production run (AMD EPYC 7742; Python 3.8.5, NumPy 1.19.2, SciPy 1.5.2); table 9. Eighteen labels and all quoted configurations are reproduced, and so is σ_{sys} wherever both runs have it; quoted values agree to $\leq 10^{-8}$ and σ_{stat} to $\leq 1.2 \times 10^{-7}$ absolute, because nonlinear optima agree only to $\sim 10^{-7}$. Two labels change, in opposite directions, for the same reason: the $p = 3$ start of the ladder of configuration `x0.45_background_Nmin8192_K2`, a gross misfit ($\chi^2 \approx 6$–8×10^4 for 178 degrees of freedom, rejected by the F-test at a probability below 10^{-100}), has two minima, one interior and a lower one with a_1 on its bound -50, and which one the multistart optimizer finds depends on floating-point details (table 10). A bound minimum is flagged and stops the ladder, so the configuration never reaches its good $p \geq 5$ fits.

- At $\Delta_\sigma = 0.1727335$ production found the interior minimum: four quotable fits, $\sigma_{\mathrm{sys}} = 0.00078$, ACCEPTED. The workstation (with its own libraries, and with the production versions in OpenBLAS and MKL builds) and an AMD EPYC 7543 node with the production versions find the bound minimum: three fits remain, σ_{sys} needs four, CONDITIONAL. The fourth fit, which exists only on the production platform, has $\chi^2/\mathrm{dof} = 1.47$ against the cap 1.5.
- At $\Delta_\sigma = 0.1625$ it is the reverse: production found the bound minimum (five quotable fits, all of variant `second`, so σ_{sys} is not available: CONDITIONAL); the workstation finds the interior minimum, a sixth fit of variant `background` becomes quotable, $\sigma_{\mathrm{sys}} = 0.00024$ and the label is ACCEPTED.

The values and σ_{stat} are robust; these two labels should be read as platform-sensitive.

6.4.2 Sensitivity to the ladder rule

To measure how much the trimmed families matter, all 20 couplings were refitted twice in one job on one node (AMD EPYC 7543; Python 3.8.5, NumPy 1.19.2, SciPy 1.5.2, MKL): once with the frozen fitter, and once with a copy whose only change is that a flagged lower-degree fit no longer stops the ladder (its χ^2 still enters the F-test). This is a diagnostic, not the protocol, and table 5 is not modified; records and the modified fitter are in `results/1d/robust_ladder_20261013/`. Table 11 gives the result. On this node the frozen ladder reproduces the workstation refit of section 6.4.1, including its two label changes. With the robust ladder:

- every quoted value is unchanged to 2×10^{-5};
- the number of quotable fits grows at eleven couplings, and σ_{sys} grows at 0.125 (0.00193 to 0.00221), 0.1375 (0.00180 to 0.00192), 0.1625 (0.00024 to 0.00071) and 0.175 (0.00067 to 0.00115);

Table 9: $d = 1$ fitting stage rerun from the released snapshots (workstation, 2026-10-07) against production. n_q: quotable fits; $|\delta\Delta_\varepsilon|$: difference of the quoted values. Records: `results/1d/refit_workstation_20261007/`.

	production			refit (workstation)				
Δ_σ	decision	n_q	σ_{sys}	decision	n_q	σ_{sys}	$\lvert\delta\Delta_\varepsilon\rvert$	
0.0125	X-CHECK UNAVAIL.	72	0.01003	X-CHECK UNAVAIL.	72	0.01003	3e-09	
0.025	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	ACCEPTED (σ_{sys}-DOM.)	68	0.00959	1e-09	
0.0375	X-CHECK UNAVAIL.	71	0.00425	X-CHECK UNAVAIL.	71	0.00425	2e-09	
0.05	ACCEPTED	72	0.00220	ACCEPTED	72	0.00220	5e-09	
0.0625	X-CHECK UNAVAIL.	50	0.00150	X-CHECK UNAVAIL.	50	0.00150	4e-09	
0.075	ACCEPTED	30	0.00139	ACCEPTED	30	0.00139	2e-09	
0.0875	ACCEPTED	34	0.00194	ACCEPTED	34	0.00194	2e-09	
0.1	ACCEPTED (σ_{sys}-DOM.)	48	0.00242	ACCEPTED (σ_{sys}-DOM.)	48	0.00242	8e-10	
0.1125	ACCEPTED (σ_{sys}-DOM.)	41	0.00217	ACCEPTED (σ_{sys}-DOM.)	41	0.00217	2e-09	
0.125	ACCEPTED	9	0.00193	ACCEPTED	9	0.00193	3e-09	
0.1375	ACCEPTED (σ_{sys}-DOM.)	17	0.00180	ACCEPTED (σ_{sys}-DOM.)	17	0.00180	2e-09	
0.15	X-CHECK FAILED	11	0.00306	X-CHECK FAILED	11	0.00306	2e-10	
0.1625	CONDITIONAL	5	–	ACCEPTED	6	0.00024	2e-09	changed
0.1727335	ACCEPTED	4	0.00078	CONDITIONAL	3	–	3e-09	changed
0.175	X-CHECK FAILED	6	0.00067	X-CHECK FAILED	6	0.00067	3e-09	
0.1875	CONDITIONAL	4	–	CONDITIONAL	4	–	6e-09	
0.2	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	
0.2125	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	
0.225	CONDITIONAL	1	–	CONDITIONAL	1	–	7e-09	
0.2375	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	

Table 10: The $p = 3$ start of the ladder of `x0.45_background_Nmin8192_K2` on each platform: χ^2 (178 dof), the field coefficient a_1 (bounds ± 50), and the consequence for the ladder.

Δ_σ	platform	χ^2	a_1	minimum	ladder
0.1625	production (AMD EPYC 7742)	60656.3	-50.00	bound	stops at $p = 3$
0.1625	Intel i9-13900K, Py 3.12/NumPy 1.26/SciPy 1.11, OpenBLAS	67496.2	-38.29	interior	continues
0.1727335	production (AMD EPYC 7742)	75400.9	-45.49	interior	continues
0.1727335	Intel i9-13900K, Py 3.12/NumPy 1.26/SciPy 1.11, OpenBLAS	68429.4	-50.00	bound	stops at $p = 3$
0.1727335	Intel i9-13900K, Py 3.8.5/NumPy 1.19.2/SciPy 1.5.2, OpenBLAS	68429.4	-50.00	bound	stops at $p = 3$
0.1727335	Intel i9-13900K, same versions, MKL	68429.4	-50.00	bound	stops at $p = 3$
0.1727335	AMD EPYC 7543, same versions, MKL	68429.4	-50.00	bound	stops at $p = 3$

- both platform-sensitive couplings pass: 0.1625 with $\sigma_{\text{sys}} = 0.00071$, and 0.1727335 with nine quotable fits and the production value $\sigma_{\text{sys}} = 0.00078$;
- 0.125 passes as σ_{sys}-dominated, 0.1875 becomes CROSS-CHECK UNAVAILABLE with $\sigma_{\text{sys}} = 0.00063$, and 0.2 becomes CONDITIONAL with two quotable fits.

The trimming therefore mostly shrinks the model families; it decides a label only where few fits are quotable. Under the robust rule ten couplings would pass, and the two platform-sensitive labels would no longer depend on which minimum the degree-3 fit finds.

6.5 Comparison with theory and earlier simulations

Resummed ε expansion. Table 12 compares with the eight-loop ε expansion of the accompanying article [1], resummed by conformal Borel resummation [3]. In $d = 1$ it is resummed in the variable $v = 1 - 2\sqrt{\Delta_\sigma}$, with the short-range series near $s = 1$ imposed; we call this curve CB8 (values and band from the article's resummation package, `data/reference/theory_1d_CB8.csv`). The band is a sensitivity range over resummation prescriptions, not a confidence interval. The nine values that passed the protocol differ from CB8 by -0.0024 to $+0.0032$, that is 1 to 13 times their σ_{stat}; four of them (0.025, 0.05, 0.1, 0.1125) lie outside the band. As a scale only, σ_{sys} exceeds the difference except at 0.05, 0.1125 and 0.1727335 (1.1, 1.5 and 1.2 times σ_{sys}). Near the Gaussian end the series needs little resummation (at $\Delta_\sigma = 0.2375$, $\varepsilon = 0.05$, its partial sums are stable to 3×10^{-4}). There the largest-pair effective dimensions (0.2, 0.2125, 0.2375) and the single quoted fit at 0.225 lie 0.0012, 0.0065, 0.0104 and 0.0090 below it. A slow approach with exponent $\omega \approx \varepsilon$, which the fit windows and the crossing grid ($\omega' \geq 0.2$) cannot resolve, would give a deficit of this sign; we have not estimated its size. These values are finite-size results.

Earlier simulations. Table 13 lists the published Monte Carlo values at our couplings (earlier estimates from exact finite-range scaling [24] are not simulations and are not listed). Published values exist at six of our couplings; at three of them (0.05, 0.1, 0.1727335) our value passed the protocol, and at $\Delta_\sigma = 0.2$ the comparison is with a finite-size value. The equilibrium values (UGA, APR, Luijten's thesis) agree with ours to within two standard deviations of the difference, computed from the published error and our σ_{stat}; σ_{sys} is not added, because it is not a standard deviation. The largest ratio, 1.98, is at $\Delta_\sigma = 0.05$, where the difference is

Table 11: $d = 1$ ladder sensitivity: production, and the frozen and the robust ladder refitted on one node. n_q: quotable fits; $|\delta\Delta_\varepsilon|$: quoted value, robust minus production; last column: what the robust ladder changes relative to the frozen ladder on the same node.

	production			frozen ladder			robust ladder				
Δ_σ	decision	n_q	$\sigma_{\rm sys}$	decision	n_q	$\sigma_{\rm sys}$	decision	n_q	$\sigma_{\rm sys}$	$\lvert\delta\Delta_\varepsilon\rvert$	
0.0125	X-CHECK UNAVAIL.	72	0.01003	X-CHECK UNAVAIL.	72	0.01003	X-CHECK UNAVAIL.	72	0.01003	1e-09	
0.025	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	68	0.00959	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	68	0.00959	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	68	0.00959	2e-10	
0.0375	X-CHECK UNAVAIL.	71	0.00425	X-CHECK UNAVAIL.	71	0.00425	X-CHECK UNAVAIL.	71	0.00425	9e-10	
0.05	ACCEPTED	72	0.00220	ACCEPTED	72	0.00220	ACCEPTED	72	0.00220	6e-09	
0.0625	X-CHECK UNAVAIL.	50	0.00150	X-CHECK UNAVAIL.	50	0.00150	X-CHECK UNAVAIL.	50	0.00150	3e-09	
0.075	ACCEPTED	30	0.00139	ACCEPTED	30	0.00139	ACCEPTED	30	0.00139	6e-09	
0.0875	ACCEPTED	34	0.00194	ACCEPTED	34	0.00194	ACCEPTED	35	0.00194	2e-09	n_q
0.1	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	48	0.00242	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	48	0.00242	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	49	0.00242	1e-10	n_q
0.1125	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	41	0.00217	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	41	0.00217	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	44	0.00217	4e-09	n_q
0.125	ACCEPTED	9	0.00193	ACCEPTED	9	0.00193	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	18	0.00221	2e-05	label
0.1375	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	17	0.00180	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	17	0.00180	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	24	0.00192	1e-09	n_q
0.15	X-CHECK FAILED	11	0.00306	X-CHECK FAILED	11	0.00306	X-CHECK FAILED	19	0.00306	1e-09	n_q
0.1625	CONDITIONAL	5	–	ACCEPTED	6	0.00024	ACCEPTED	15	0.00071	4e-09	n_q
0.1727335	ACCEPTED	4	0.00078	CONDITIONAL	3	–	ACCEPTED	9	0.00078	2e-09	label
0.175	X-CHECK FAILED	6	0.00067	X-CHECK FAILED	6	0.00067	X-CHECK FAILED	13	0.00115	5e-11	n_q
0.1875	CONDITIONAL	4	–	CONDITIONAL	4	–	X-CHECK UNAVAIL.	10	0.00063	2e-09	label
0.2	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	CONDITIONAL	2	–	–	label
0.2125	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	
0.225	CONDITIONAL	1	–	CONDITIONAL	1	–	CONDITIONAL	1	–	5e-09	
0.2375	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	MODEL-LIMITED	0	–	–	

Table 12: $d = 1$ against the resummed ε expansion (CB8, with band). MC: the quoted global fit ($\sigma_{\rm stat}$ only), or the largest-pair $\Delta_{\rm eff}(U)$ where no fit is quotable.

Δ_σ	ε	CB8	band	decision	MC kind	MC	MC − CB8
0.0125	0.950	0.7892	[0.7891, 0.7892]	X-CHECK UNAVAIL.	fit	0.78404(77)	-0.0051
0.025	0.900	0.7134	[0.7133, 0.7136]	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	fit	0.71115(49)	-0.0023
0.0375	0.850	0.6615	[0.6611, 0.6619]	X-CHECK UNAVAIL.	fit	0.65904(51)	-0.0024
0.05	0.800	0.6222	[0.6215, 0.6230]	ACCEPTED	fit	0.61987(55)	-0.0024
0.0625	0.750	0.5914	[0.5903, 0.5924]	X-CHECK UNAVAIL.	fit	0.59003(35)	-0.0014
0.075	0.700	0.5666	[0.5650, 0.5679]	ACCEPTED	fit	0.56694(37)	+0.0003
0.0875	0.650	0.5466	[0.5446, 0.5479]	ACCEPTED	fit	0.54761(32)	+0.0011
0.1	0.600	0.5302	[0.5280, 0.5316]	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	fit	0.53248(20)	+0.0022
0.1125	0.550	0.5170	[0.5146, 0.5188]	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	fit	0.52020(25)	+0.0032
0.125	0.500	0.5064	[0.5040, 0.5086]	ACCEPTED	fit	0.50710(35)	+0.0007
0.1375	0.450	0.4981	[0.4958, 0.5005]	ACCEPTED ($\sigma_{\rm sys}$-DOM.)	fit	0.49856(30)	+0.0005
0.15	0.400	0.4917	[0.4898, 0.4942]	X-CHECK FAILED	fit	0.49325(41)	+0.0015
0.1625	0.350	0.4871	[0.4857, 0.4895]	CONDITIONAL	fit	0.48761(58)	+0.0005
0.1727335	0.309	0.4846	[0.4835, 0.4867]	ACCEPTED	fit	0.48368(52)	-0.0010
0.175	0.300	0.4842	[0.4832, 0.4862]	X-CHECK FAILED	fit	0.48460(35)	+0.0004
0.1875	0.250	0.4829	[0.4822, 0.4844]	CONDITIONAL	fit	0.48263(30)	-0.0003
0.2	0.200	0.4831	[0.4828, 0.4840]	MODEL-LIMITED	largest pair	0.4819(15)	-0.0012
0.2125	0.150	0.4848	[0.4847, 0.4852]	MODEL-LIMITED	largest pair	0.4784(15)	-0.0065
0.225	0.100	0.4881	[0.4881, 0.4882]	CONDITIONAL	fit	0.47912(68)	-0.0090
0.2375	0.050	0.4930	[0.4930, 0.4930]	MODEL-LIMITED	largest pair	0.4826(15)	-0.0104

nine times our σ_{sys}. The nonequilibrium-relaxation values of Tomita lie below; his own estimate of η misses the exact $\eta = 2 - s$ by 2.5–3.5 quoted errors at $s = 0.6$–0.8, so his quoted errors do not cover the systematic errors of that method, and the disagreement at $\Delta_\sigma = 0.1$ is with an estimate that fails an exact check. APR's correction exponent $\omega = 0.201(11)$ at $\Delta_\sigma = 0.1727335$ agrees with our fitted $\omega = 0.195$ (table 7).

Table 13: Published $d = 1$ values ($\Delta_\varepsilon = 1 - 1/\nu$ where ν is quoted) against ours. UGA: Uzelac–Glumac–Aničić; APR: Angelini–Parisi–Ricci-Tersenghi; NER: nonequilibrium relaxation. Ours: value(σ_{stat})(σ_{sys}), or a finite-size value.

Δ_σ	source	method	published	ours	our label	published − ours
0.05	Tomita 2009 [23]	MC, NER	0.5633(362)	0.61987(55)(220)	ACCEPTED	-0.0566
0.05	UGA 2001 [19]	MC, largest cluster	0.6000(100)	0.61987(55)(220)	ACCEPTED	-0.0199
0.0625	APR 2014 [20]	MC, FSS	0.5876(13)	0.59003(35)(150)	X-CHECK UNAVAIL.	-0.0024
0.1	Luijten 1997 (thesis) [21]	MC, FSS	0.5392(106)	0.53248(20)(242)	ACCEPTED (σ_{sys}-DOM.)	+0.0067
0.1	Tomita 2009 [23]	MC, NER	0.4709(140)	0.53248(20)(242)	ACCEPTED (σ_{sys}-DOM.)	-0.0616
0.1	UGA 2001 [19]	MC, largest cluster	0.5300(200)	0.53248(20)(242)	ACCEPTED (σ_{sys}-DOM.)	-0.0025
0.15	Tomita 2009 [23]	MC, NER	0.4536(508)	0.49325(41)(306)	X-CHECK FAILED	-0.0397
0.15	UGA 2001 [19]	MC, largest cluster	0.4900(200)	0.49325(41)(306)	X-CHECK FAILED	-0.0033
0.1727335	APR 2014 [20]	MC, FSS	0.4940(140)	0.48368(52)(78)	ACCEPTED	+0.0103
0.2	Tomita 2009 [23]	MC, NER	0.4444(648)	0.4819(15) [finite size]	MODEL-LIMITED	-0.0375
0.2	UGA 2001 [19]	MC, largest cluster	0.5000(200)	0.4819(15) [finite size]	MODEL-LIMITED	+0.0181

6.6 Status of the design goals

The campaign had three design goals: ACCEPTED at every $\Delta_\sigma \in [0.0625, 0.1875]$, which was not met (7 of 12 couplings); a definite label at every outer coupling, which was met; and exact reproduction of the two-point campaign's values in its original windows, which was met. An extension to $N = 2^{20}$ (called N3) was not run.

7 $d = 2$ and $d = 3$: common protocol

7.1 Design and analysis

For each coupling, five sizes form four size-doubling pairs, the minimum for an extrapolation with a free exponent. Temperatures are centered on crossings measured in a short stage-0 campaign (frozen tables `data/2d/production_centers.csv`, `data/3d/production_centers_3d.csv`), five per size. Every (L, β) group has 12 hot/cold replicas with 10^6 measured sweeps; each dimension has 2 400 chains.

1. For each pair and $Q \in \{U_4, U_6, U_8, Y\}$ the effective dimension (4) is computed at the U_4 crossing; the bootstrap gives the covariance over pairs and observables.
2. Each series $\Delta^Q_{\text{eff}}(L)$ is extrapolated as $\Delta_\varepsilon + c_Q L^{-\omega'}$ by generalized least squares, with ω' profiled on $[0.2, 2]$ (step 0.01); a *joint* series fits U_4 and Y with common Δ_ε and ω' and separate amplitudes (written U4+Y).
3. A series is eligible if $\chi^2/\text{dof} \le 2$, its $\Delta\chi^2 \le 1$ interval of ω' lies inside $[0.22, 1.98]$, and its σ_∞ is at most 5 times the error of the largest pair.
4. The quoted value is that of the joint series if eligible, else U_4, else the first eligible one of U_6, U_8, Y; σ_{sys} is half the range of Δ_ε over the eligible series plus half the range of the quoted series' Δ_ε over its $\Delta\chi^2 \le 1$ interval of ω'.
5. Cross-check: the extrapolated U_4 and Y must agree within $2\sigma_D$; reblocking at blocks 1000 and 4000 against 2000.

The grid floor $\omega' \ge 0.2$ excludes the correction exponent expected near the Gaussian end ($\omega \approx \varepsilon = 0.1$ at 2D $s = 1.05$ and 3D $s = 1.55$), and near the short-range end the relevant exponent $s_* - s$ is also below 0.2; there the labels partly reflect this design choice.

7.2 The statistical error of an extrapolation: amendment P1

The frozen protocols compute σ_{stat}, σ_D and the reblocking comparison by refitting each bootstrap draw at the *central* ω', i.e. conditionally on a fitted exponent. A review of the finished evaluations identified this. Amendment P1 re-profiles ω' in every draw (same grid and covariance; draws on a grid edge are kept). It leaves the eligibility gates, σ_{sys}, the choice of series and the decision order unchanged. The gate $\sigma_\infty \le 5\times$ (error of the largest pair) is therefore still evaluated with the fixed-ω' σ_∞. With the profiled one, three series would fail it: Y at 2D $s = 1.05$ and U_4 at 3D $s = 1.55$ and 1.7. 2D $s = 1.05$ would then have no eligible series,

and the 3D cross-check would become unavailable at 1.55 and 1.7. The series of the accepted value at 3D $s = 1.6$ pass the gate either way.

P1 was calibrated before adoption on synthetic data whose truths are the recorded fits: for every coupling, single-series ensembles for each of U_4, U_6, U_8, Y and a joint ensemble, 2 000 synthetic data sets each with 500 synthetic draws, run through the full fit, gates and both error rules. A coupling is evaluated under P1 only if every series it uses has profiled 1σ coverage ≥ 0.60 and 2σ coverage ≥ 0.90 (with at least 200 eligible fits) and, when both U_4 and Y are eligible, the profiled cross-check's false-failure rate is ≤ 0.10; otherwise the frozen rule applies (appendix C).

Results:

- The fixed-ω' rule under-covers mainly for single series: in the strata that pass the adoption rules its 1σ coverage is 0.00–0.38 for U_4 and 0.12–0.29 for Y, against 0.59–0.67 for the joint series (pooled over all strata, the 1σ coverage is 0.287 for 45 440 eligible synthetic fits). In the same strata the profiled rule covers 0.81–1.00 for the Binder series, 0.69–0.83 for Y and 0.67–0.70 for the joint series.
- The false-failure rate of the cross-check at 3D $s = 1.6$ is 0.7 % profiled against 61 % fixed.

The calibration is *conditional on the single-power model*. We tested one misspecified family, which adds an $L^{-2\omega'}$ term that reproduces the recorded $U_4 - Y$ difference. In it the joint series at 3D $s = 1.6$ is biased by -0.0062 with profiled coverage 0, and the profiled cross-check does not detect the bias. But U_4 is then ineligible in every synthetic set, so that coupling could not have been accepted under this misspecification. The calibration therefore does not test model errors; the one misspecification studied would have blocked the label rather than biased it. Exactly one label changes under P1: 3D $s = 1.6$, from CROSS-CHECK FAILED to ACCEPTED. This was predicted before implementation, but after the frozen $z = -2.87$ was known.

8 $d = 2$

Couplings $s \in \{1.05, \ldots, 1.7\}$ ($\Delta_\sigma = 0.475 \ldots 0.15$), sizes $L = 64 \ldots 1024$, temperature half-width $w(L) = 0.0015(L/64)^{-0.75}$. Stage 0 used $L \leq 512$; the preflight confirmed every $512 \to 1024$ crossing inside its bracket. The production ran on 2026-09-20 to 22 ($L = 1024$ at 88 ms per sweep, 25–31 h per chain).

Reading. No 2D coupling has a value that passed the protocol. The robust statement is finite-size: $\Delta_{\text{eff}}(U_4, 512 \to 1024) = 0.958$–$0.977$ at all eight couplings ($\sigma \approx 0.002$), moving by ≤ 0.01 over the four pairs and always below the endpoint value 1. Against the resummed expansion (table 18) the largest-pair U_4 agrees within 0.0015 for $s = 1.2$–1.4 and falls short by 0.005–0.02 towards both ends. At $s = 1.2$, 1.3 and 1.4 the other ways of imposing the end point move the expansion by up to 0.0005, 0.0023 and 0.0047, so there the agreement is as good as the expansion is determined. Y approaches U_4 as L doubles at every coupling, slowly and from below at $s \leq 1.2$ and from above at $s \geq 1.4$; near s_* this is consistent with the slow crossover of section 2. A single power $L^{-\omega'}$ with $\omega' \in [0.2, 2]$ is either unresolved or inconsistent over $L = 64$–1024 (appendix B). The three CONDITIONAL values rest on one series without σ_{sys}; at $s = 1.05$ and 1.1 that series is Y, and its extrapolations (0.921, 0.928) lie 0.065 and 0.047 below the expansion, 3.7 and 4.6 times their σ_{stat}. **These two values should not be used as infinite-size estimates.** The crossing inverse temperatures $\beta_\times$ still move by up to 1.7×10^{-5} between the last two pairs; they are finite-size values.

9 $d = 3$

Couplings $s \in \{1.55, \ldots, 1.95\}$ ($\Delta_\sigma = 0.725 \ldots 0.525$), sizes $L = 8 \ldots 128$, half-width $w(L) = 10^{-3}(L/16)^{-1}$. An $L = 128$ chain runs at 105 ms per sweep alone but at about 250 ms when four chains share a 4-core CCX (a group of cores sharing a 16 MB L3 cache); the production therefore pinned one $L = 128$ chain per CCX (32 per 128-core node, ≈ 30 h per chain). The preflight gate and its amendment are described in section 5. The production ran on 2026-09-21 to 23.

The value at $s = 1.6$. At $s = 1.6$ ($\Delta_\sigma = 0.70$) all five series are eligible: U_4 1.4726, U_6 1.4720, U_8 1.4709, Y 1.4750, U4+Y 1.4731. The quoted value comes from the joint series: $\Delta_\varepsilon = 1.47314 \pm 0.00118\,(\text{stat}) \pm 0.00230\,(\text{sys})$ ($\omega' = 1.24$, $\chi^2 = 2.38/4$); the range over the U_4 and Y series alone is 1.4726–1.4750. It passed the protocol only under amendment P1, and the precision suffix remains ($\sigma = 0.0031 > 0.003$ at $64 \to 128$). Three facts limit what the label shows. The cross-check has little power here: the extrapolated $Y - U_4$ changes sign between $s = 1.55$ and 1.7 (-0.033, $+0.002$, $+0.068$), so their agreement at 1.6 is partly a crossing of two biased estimates, and the profiled $\sigma_D \approx 0.0055$ cannot test a σ_{stat} of 0.0012. The fitted $\omega' = 1.24$ is far above the leading exponents expected at $s = 1.6$ ($\omega \approx \varepsilon \approx 0.2$; $s_* - s \approx 0.36$), so the single power is an effective

Table 14: $d = 2$ finite-size data (block 2000): for each doubling pair the crossing inverse temperature $\beta_\times$ (error in units of 10^{-7}) and $\Delta_{\text{eff}}(U_4)$, $\Delta_{\text{eff}}(Y)$ (errors in units of 10^{-4}, 500 bootstrap draws).

		$64 \to 128$			$128 \to 256$			$256 \to 512$			$512 \to 1024$		
s	Δ_σ	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y
1.05	0.475	0.1300733(16)	0.9633(13)	0.8390(14)	0.1300298(10)	0.9661(17)	0.8658(18)	0.1300145(6)	0.9708(20)	0.8863(21)	0.1300084(4)	0.9702(25)	0.8964(23)
1.1	0.45	0.1355249(16)	0.9609(14)	0.8608(15)	0.1354801(10)	0.9639(17)	0.8912(18)	0.1354691(6)	0.9664(20)	0.9045(19)	0.1354627(4)	0.9659(23)	0.9172(23)
1.2	0.4	0.1463564(17)	0.9610(12)	0.9142(13)	0.1463343(10)	0.9576(16)	0.9281(16)	0.1463269(5)	0.9616(19)	0.9391(19)	0.1463241(3)	0.9631(22)	0.9478(22)
1.3	0.35	0.1571230(17)	0.9581(12)	0.9561(12)	0.1571157(9)	0.9586(15)	0.9683(15)	0.1571170(6)	0.9605(17)	0.9709(18)	0.1571177(3)	0.9584(20)	0.9698(20)
1.4	0.3	0.1677986(17)	0.9599(12)	1.0024(12)	0.1678175(9)	0.9590(14)	1.0032(14)	0.1678263(5)	0.9582(17)	0.9986(16)	0.1678310(3)	0.9627(20)	0.9977(19)
1.5	0.25	0.1783705(17)	0.9608(12)	1.0484(11)	0.1784129(10)	0.9615(14)	1.0423(13)	0.1784331(5)	0.9640(16)	1.0339(15)	0.1784417(3)	0.9661(18)	1.0245(17)
1.6	0.2	0.1888074(18)	0.9653(12)	1.0977(11)	0.1888717(10)	0.9652(13)	1.0885(12)	0.1889039(5)	0.9691(15)	1.0729(13)	0.1889149(3)	0.9698(18)	1.0620(15)
1.7	0.15	0.1990663(19)	0.9671(11)	1.1516(10)	0.1991553(10)	0.9699(13)	1.1381(11)	0.1991952(5)	0.9751(16)	1.1254(14)	0.1992121(3)	0.9773(17)	1.1088(14)

Table 15: $d = 2$ labels under the frozen rule and under P1 (with the rule actually applied), the quoted value, $\sigma_{\rm stat}$ under both rules, $\sigma_{\rm sys}$, and the $U_4 - Y$ cross-check (z with the fixed and the profiled σ_D). The CONDITIONAL values at $s = 1.05$ and 1.1 are single-series (Y) extrapolations 0.065 and 0.047 below the resummed expansion (table 18); they are not infinite-size estimates.

s	Δ_σ	frozen label	P1 label	rule	Δ_ε	$\sigma_{\rm stat}$ frozen	$\sigma_{\rm stat}$ P1	$\sigma_{\rm sys}$	$U_4 - Y$; z fixed / profiled
1.05	0.475	CONDITIONAL	CONDITIONAL	profiled	0.92078	0.00181	0.01773	–	–
1.1	0.45	CONDITIONAL	CONDITIONAL	profiled	0.92846	0.00145	0.01012	–	–
1.2	0.4	CONDITIONAL	CONDITIONAL	profiled	0.96202	0.00162	0.00182	–	–
1.3	0.35	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.4	0.3	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.5	0.25	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.6	0.2	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.7	0.15	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–

description. And the label changed under an amendment adopted after the frozen result was known. The value is *consistent* with the resummed expansion (1.4721) and with the finite-size $\Delta_{\rm eff}(U_4, 64 \to 128) = 1.4715(31)$, which is the main reason to trust it.

Reading. The finite-size U_4 exponent at $64 \to 128$ decreases from 1.473 ($s = 1.55$) to 1.449 ($s = 1.95$), between the Gaussian value 1.5 and the short-range value 1.4126. Y does not converge towards U_4 at $s \geq 1.7$: it lies above U_4 from $16 \to 32$ on and keeps rising (1.53–1.68 at $64 \to 128$). Against the resummed expansion (table 18), the extrapolated values at $s = 1.7$–1.9 lie 0.022–0.033 above. At 1.7 and 1.85 the label is CROSS-CHECK FAILED, and the differences (+0.022, +0.030) are smaller than $\sigma_{\rm sys}$ (0.052, 0.091), which is large because the series disagree; at 1.9 (CROSS-CHECK UNAVAILABLE, $\sigma_{\rm sys} = 0.0045$) the difference is 4.6 times $\sigma_{\rm stat}$. Near s_* the U_4 series therefore sits high, and the Y–U_4 split matches in size the slow crossover of section 2 at the largest pair, although its growth with L up to 128 is not reproduced by that leading-order picture.

9.1 $d = 2, 3$ against theory and earlier work

Table 18 compares with CB8 and figure 2 shows the effective dimensions. Table 19 lists published values inside the window. In $d = 2$ the Monte Carlo values (APR, Luijten's thesis, Picco) have errors 0.02–0.034 and differ from our largest-pair $\Delta_{\rm eff}(U_4)$ by at most two of their errors; they do not distinguish our finite-size values from CB8. In $d = 3$, at $s = 1.6$, Luijten's $L \leq 64$ values (1.477(16), 1.500(20)) and the functional-RG value (1.475) lie 0.002–0.027 above our accepted value, within 1.3 of their errors. Nearer s_* the bootstrap values agree with CB8 to about 0.01 or lie slightly below it, while the functional-RG values are close to our largest-pair $\Delta_{\rm eff}(U_4)$, up to 0.03 above CB8 (both read off figures). Published values at the ends of the window, $s = d/2$ in $d = 3$ [22] and $s = s_*$ in $d = 2$ [26], are in the reference files but outside our grid.

Table 16: $d = 3$ finite-size data (block 2000), as table 14.

		$8 \to 16$			$16 \to 32$			$32 \to 64$			$64 \to 128$		
s	Δ_σ	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y	$\beta_\times$	U_4	Y
1.55	0.725	0.0894786(14)	1.4427(11)	1.2535(12)	0.0893412(7)	1.4583(15)	1.3545(16)	0.0893146(4)	1.4677(22)	1.4028(23)	0.0893087(2)	1.4729(30)	1.4269(30)
1.6	0.7	0.0916831(15)	1.4398(11)	1.2914(13)	0.0915467(8)	1.4596(15)	1.3970(17)	0.0915234(4)	1.4661(21)	1.4382(23)	0.0915178(2)	1.4715(31)	1.4628(30)
1.7	0.65	0.0959967(15)	1.4411(11)	1.3761(12)	0.0958738(8)	1.4528(16)	1.4673(17)	0.0958528(3)	1.4612(22)	1.5085(21)	0.0958497(2)	1.4650(29)	1.5263(29)
1.75	0.625	0.0981147(16)	1.4377(11)	1.4115(12)	0.0979933(8)	1.4535(15)	1.5071(16)	0.0979751(3)	1.4536(21)	1.5379(21)	0.0979722(2)	1.4615(27)	1.5596(27)
1.8	0.6	0.1002020(16)	1.4372(11)	1.4513(12)	0.1000841(8)	1.4505(14)	1.5417(15)	0.1000677(3)	1.4563(20)	1.5795(20)	0.1000665(2)	1.4515(27)	1.5818(27)
1.85	0.575	0.1022550(15)	1.4364(11)	1.4929(12)	0.1021468(8)	1.4490(15)	1.5776(16)	0.1021325(3)	1.4538(20)	1.6113(20)	0.1021317(2)	1.4569(27)	1.6221(27)
1.9	0.55	0.1042865(17)	1.4370(10)	1.5325(12)	0.1041795(8)	1.4458(14)	1.6148(15)	0.1041684(4)	1.4526(19)	1.6464(20)	0.1041687(2)	1.4511(26)	1.6488(25)
1.95	0.525	0.1062845(16)	1.4320(11)	1.5694(11)	0.1061853(7)	1.4466(15)	1.6539(15)	0.1061755(4)	1.4481(20)	1.6771(21)	0.1061771(2)	1.4493(28)	1.6790(27)

Table 17: $d = 3$ labels, as table 15. At $s = 1.7$ and 1.85 P1 is not calibrated (rule "fixed", appendix C): the profiled z is shown for information only, and the label uses the fixed z. The extrapolated values at $s = 1.7$–1.9 lie 0.022–0.033 above the resummed expansion (table 18).

s	Δ_σ	frozen label	P1 label	rule	Δ_ε	σ_{stat} frozen	σ_{stat} P1	σ_{sys}	$U_4 - Y$; z fixed / profiled
1.55	0.725	X-CHECK FAILED [σ]	X-CHECK FAILED [σ]	profiled	1.48118	0.00165	0.01632	0.02716	+0.0326; +29.48 / +2.49
1.6	0.7	X-CHECK FAILED [σ]	ACCEPTED [σ]	profiled	1.47314	0.00112	0.00118	0.00230	-0.0025; -2.87 / -0.45
1.7	0.65	X-CHECK FAILED	X-CHECK FAILED	fixed	1.47375	0.00174	0.00174	0.05237	-0.0676; -56.57 / -4.90
1.75	0.625	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.8	0.6	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–
1.85	0.575	X-CHECK FAILED	X-CHECK FAILED	fixed	1.45830	0.00103	0.00103	0.09064	-0.1723; -232.64 / -28.05
1.9	0.55	X-CHECK UNAVAIL.	X-CHECK UNAVAIL.	profiled	1.45449	0.00113	0.00720	0.00452	–
1.95	0.525	MODEL-LIMITED	MODEL-LIMITED	fixed	–	–	–	–	–

Table 18: $d = 2$ (top) and $d = 3$ (bottom) against the resummed ε expansion of the accompanying article (CB8, with band; `data/reference/theory_2d3d_CB8.csv`). CB8 imposes the short-range end: in $d = 2$ the value 1, zero slope and the curvature, in $d = 3$ the value and the slope 0.27315, itself an estimate (range 0.269–0.276). The band is a sensitivity range over prescriptions; it excludes the way the end point is imposed, which in $d = 2$ moves the curve by up to 0.0005, 0.0023, 0.0047, 0.0057, 0.0070 and 0.0055 at $s = 1.2, \ldots, 1.7$, and the 3D slope range. Last two columns: quoted value minus CB8, and largest-pair $\Delta_{\text{eff}}(U_4)$ minus CB8.

s	P1 label	Δ_ε	σ_{stat}	σ_{sys}	$\Delta_{\text{eff}}(U_4)$ largest pair	CB8	band	$\Delta_\varepsilon - $CB8	$U_4 - $CB8
1.05	CONDITIONAL	0.92078	0.01773	–	0.9702(25)	0.9858	[0.9858, 0.9858]	-0.0651	-0.0157
1.1	CONDITIONAL	0.92846	0.01012	–	0.9659(23)	0.9754	[0.9753, 0.9754]	-0.0470	-0.0095
1.2	CONDITIONAL	0.96202	0.00182	–	0.9631(22)	0.9626	[0.9609, 0.9628]	-0.0005	+0.0006
1.3	MODEL-LIMITED	–	–	–	0.9584(20)	0.9581	[0.9546, 0.9589]	–	+0.0003
1.4	MODEL-LIMITED	–	–	–	0.9627(20)	0.9612	[0.9572, 0.9624]	–	+0.0015
1.5	MODEL-LIMITED	–	–	–	0.9661(18)	0.9711	[0.9684, 0.9722]	–	-0.0050
1.6	MODEL-LIMITED	–	–	–	0.9698(18)	0.9854	[0.9845, 0.9858]	–	-0.0157
1.7	MODEL-LIMITED	–	–	–	0.9773(17)	0.9978	[0.9977, 0.9978]	–	-0.0205
1.55	X-CHECK FAILED [σ]	1.48118	0.01632	0.02716	1.4729(30)	1.4849	[1.4849, 1.4849]	-0.0037	-0.0119
1.6	ACCEPTED [σ]	1.47314	0.00118	0.00230	1.4715(31)	1.4721	[1.4721, 1.4721]	+0.0010	-0.0006
1.7	X-CHECK FAILED	1.47375	0.00174	0.05237	1.4650(29)	1.4515	[1.4515, 1.4516]	+0.0222	+0.0134
1.75	MODEL-LIMITED	–	–	–	1.4615(27)	1.4430	[1.4429, 1.4431]	–	+0.0185
1.8	MODEL-LIMITED	–	–	–	1.4515(27)	1.4352	[1.4352, 1.4353]	–	+0.0163
1.85	X-CHECK FAILED	1.45830	0.00103	0.09064	1.4569(27)	1.4281	[1.4280, 1.4281]	+0.0302	+0.0288
1.9	X-CHECK UNAVAIL.	1.45449	0.00720	0.00452	1.4511(26)	1.4212	[1.4212, 1.4213]	+0.0333	+0.0299
1.95	MODEL-LIMITED	–	–	–	1.4493(28)	1.4145	[1.4145, 1.4145]	–	+0.0348

Table 19: Published values inside the long-range window, $d = 2$ (top) and $d = 3$ (bottom), with our largest-pair $\Delta_{\text{eff}}(U_4)$ and CB8 where the coupling is on our grid. Bootstrap and functional-RG values are read off figures.

s	source	method	$L_{\max}$	Δ_ε	our $\Delta_{\text{eff}}(U_4)$	CB8
1.2	APR 2014 [20]	Monte Carlo	–	0.976(34)	0.9631(22)	0.9626
1.2	Luijten thesis 1997 [21], Tab. 5.7	Monte Carlo	400	1.000(20)	0.9631(22)	0.9626
1.2	Luijten thesis 1997 [21], Tab. 5.8	Monte Carlo	400	0.980(30)	0.9631(22)	0.9626
1.30906	APR 2014 [20]	Monte Carlo	–	0.986(33)	–	–
1.4	Luijten thesis 1997 [21], Tab. 5.7	Monte Carlo	400	1.000(20)	0.9627(20)	0.9612
1.4	Luijten thesis 1997 [21], Tab. 5.8	Monte Carlo	400	0.960(30)	0.9627(20)	0.9612
1.6	APR 2014 [20]	Monte Carlo	–	1.004(33)	0.9698(18)	0.9854
1.6	Luijten thesis 1997 [21], Tab. 5.7	Monte Carlo	400	1.020(30)	0.9698(18)	0.9854
1.6	Luijten thesis 1997 [21], Tab. 5.8	Monte Carlo	400	0.970(20)	0.9698(18)	0.9854
1.6	Picco 2012 [25]	Monte Carlo	5120	0.958(22)	0.9698(18)	0.9854
1.57	Behan 2019 [27]	bootstrap (figure-read)	–	1.484(10)	–	–
1.59274	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.470(5)	–	–
1.6	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.475(5)	1.4715(31)	1.4721
1.6	Luijten thesis 1997 [21], Tab. 5.12	Monte Carlo	64	1.477(16)	1.4715(31)	1.4721
1.6	Luijten thesis 1997 [21], Sec. 5.5.2	Monte Carlo	64	1.500(20)	1.4715(31)	1.4721
1.615	Behan 2019 [27]	bootstrap (figure-read)	–	1.468(10)	–	–
1.66	Behan 2019 [27]	bootstrap (figure-read)	–	1.461(10)	–	–
1.68548	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.450(5)	–	–
1.7	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.459(5)	1.4650(29)	1.4515
1.704	Behan 2019 [27]	bootstrap (figure-read)	–	1.451(10)	–	–
1.77822	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.432(5)	–	–
1.8	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.452(5)	1.4515(27)	1.4352
1.8	Luijten thesis 1997 [21], Tab. 5.12	Monte Carlo	64	1.460(40)	1.4515(27)	1.4352
1.87096	Behan et al. 2024 [28]	bootstrap (figure-read)	–	1.415(5)	–	–
1.9	Defenu et al. 2015 [29]	functional RG (figure-read)	–	1.453(5)	1.4511(26)	1.4212

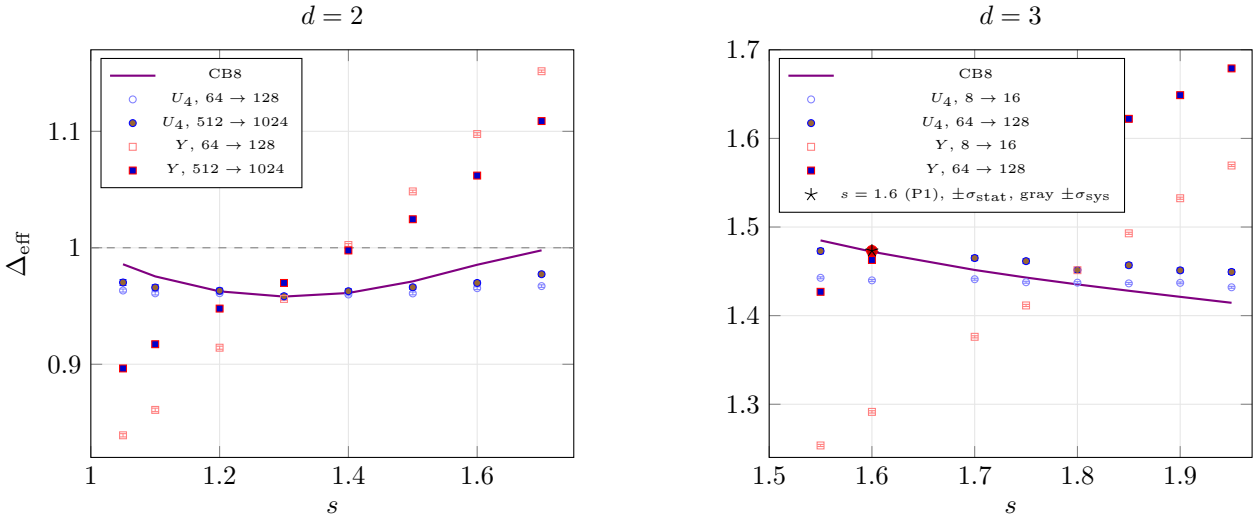


Figure 2: Effective dimensions at the smallest and the largest doubling pair, with the resummed ε expansion (CB8). Left, $d = 2$ (dashed: $\Delta_\varepsilon = 1$ at both ends of the window). Right, $d = 3$; the star is the value at $s = 1.6$ that passed the protocol under P1, with σ_{stat} (bar) and σ_{sys} (gray band) shown separately. At the largest pair the Y points (red) and the U_4 points (blue) agree in the middle of the window and differ near its ends. In $d = 2$ the difference shrinks as L doubles at every coupling; in $d = 3$ it shrinks at $s = 1.55$ and 1.6 and grows at $s \geq 1.7$.

10 Synthesis

1. In $d = 1$ nine couplings pass the protocol, with $\sigma_{\text{stat}} = (2\text{–}5.5) \times 10^{-4}$; they differ from the resummed ε expansion by -0.0024 to $+0.0032$. σ_{sys} is a sensitivity scale. The audit supports the values at 0.1, 0.1375 and 0.1727335, which are stable under stronger size cuts. It flags 0.025 (a model outside the family) and 0.175 and 0.225 (stronger size cuts). The labels at 0.1625 and 0.1727335 are platform-sensitive; two quoted fits have $P_{\chi^2} \approx 5 \times 10^{-5}$; for $\Delta_\sigma \geq 0.0875$ the global fits lie above the crossing extrapolations by a common 0.0010(3).
2. In $d = 2, 3$ the finite-size data are precise ($\sigma(\Delta_{\text{eff}}^{U_4}) \approx 0.0017$–$0.0031$ at the largest pair) and the bottleneck is the extrapolation: four doubling pairs over a factor 16 in L do not determine the correction sector, especially near the ends of the window, where the expected correction exponents are below the grid floor. More statistics at the same sizes would sharpen, not resolve, this.
3. One 3D value passes, at $s = 1.6$, under the post hoc amendment P1, and agrees with the expansion.
4. With a fitted correction exponent, a bootstrap at fixed exponent under-covers badly for single series; re-profiling the exponent in every draw restores coverage, conditionally on the extrapolation model.
5. The published equilibrium Monte Carlo values agree with ours within about two of their errors (in $d = 1$ computed with σ_{stat}, not σ_{sys}); quantitative statements should use σ_{stat} and σ_{sys} separately.

11 Open issues and next steps

1. **1D uncertainties:** a pre-declared broader model family (a U background, two correction powers, stronger size cuts), asymmetric model ranges, a converged ω' minimization and a ladder rule that does not stop on a flagged lower-degree fit, followed by a full rerun of the selection. Priority couplings: 0.025, 0.175, 0.225. A label-stability study (seeds, ladder rules, grid steps) belongs here.
2. **1D goodness of fit:** a p-value gate or a scale factor instead of $\chi^2/\text{dof} \leq 1.5$.
3. **Near the Gaussian end** in all dimensions: a crossover form $D(L) = 1 + CL^{-\varepsilon}$ with free amplitudes, or ω fixed at its perturbative value, instead of a grid with floor 0.2.
4. **Near the short-range end** ($d = 2, 3$): the crossover flow of section 2 as a model for the Y–U_4 split; measuring the effective η at the crossings would test it.
5. **Larger lattices**, only where they answer a question: 2D $L = 2048$ at $s = 1.2$ and 1.5; 3D $L = 256$ at $s = 1.9$ (about 1.5×10^4 core-hours), to tell an ordinary correction power from the slow short-range crossover by whether the Y–U_4 split grows or shrinks at $128 \to 256$.
6. **Critical inverse temperatures:** extrapolate $\beta_\times(L)$ with errors from the existing bootstrap draws.
7. **Uninspected code:** the last fixes to the P1 implementation and to the 1D tooling that records which campaigns make up each coupling (the registry).

12 Reproduction

12.1 The released folder

This note is released with the code and the records in the `MC/` folder of the public repository (code under the MIT license, everything else under CC BY 4.0, except the delivered 2D/3D engine, which keeps its own MIT license); paths below are relative to `MC/`. `README.md` states what is computed and its status; `REPRODUCE.md` gives the steps in order (environment, tests, replay, smoke run, production, merge, checks) and the decisions that a reproduction must not take on its own.

`code/1d/` 1D engines (`src/`, `src_1d/`), campaign tools, analysis (`analyze.py`, `merge_summaries.py`, `equilibration.py`, `union_check.py`), fitting (`select_model_v3.py`, `select_model.py`, `fit_fss.py`, `quotients.py`), the 20-point report (`scan_v3_report.py`) and tests.

`code/2d3d_engine/` the $d = 2, 3$ sampler and its analysis (`pack_histograms.py`, `multihist.py`, `precision_analysis.py`) and tests.

`code/2d/, code/3d/` the evaluators (frozen rule and P1) and planners.

`code/reanalysis_p1/` `calibrate_p1.py` and the tests of amendment P1.

`code/package_tools/` workspace assembly, the output comparison and the generator of this note's tables.

`protocols/` the analysis protocols, the amendment and their history (`PROTOCOL_HISTORY.md`; appendix D).

`data/` the merged production records (`DATA.md`): 1D per-coupling selected fit runs, snapshot manifests, completion evidence and registries; 2D/3D analysis records at three block sizes, campaign manifests, histogram indices and the pinned production evaluations; reference values (`data/reference/`).

`results/` expected outputs: the P1 evaluations, the P1 calibration (an input of the P1 evaluation), the 1D tables and figures, and the cited diagnostics (cross-platform refits, the robust-ladder study with its modified fitter, the cross-check study).

`slurm/` submission templates with placeholders for account, partition and paths.

The evaluators bind themselves, the calibration script and the protocols by SHA-256 and locate their inputs relative to the layout they ran in; the make targets therefore assemble that layout in a hidden `.work/` directory (code copied byte for byte, data linked), so every check passes without any edit. The production records are released byte-identical, with one exception: 25 records (from the $d = 1$ couplings $\Delta_\sigma = 0.225$ and 0.2375, and the P1 calibration) named a personal machine and were scrubbed after the results; only host labels, a home directory and file names were replaced, and the hashes inside them re-pinned. `RE-PINNED.json` lists their original and published SHA-256; the comparison maps one to the other. The other records name the cluster paths, user and compute nodes on which they were written.

12.2 How to run

Requirements: Python with NumPy and SciPy (production: Python 3.8.5, NumPy 1.19.2, SciPy 1.5.2 for the 1D fits; Python 3.8.8, NumPy 1.20.1 for the P1 calibration and evaluations; also tested with 3.12.3, 1.26.4, 1.11.4), pytest, matplotlib, a C++17 compiler and GNU make for the engines, and pdflatex for this note.

target	runs	reproduces
`test`	builds the engines; 1D protocol tests; the tests of amendment P1	the verification claims of section 7.2
`replay`	everything below that needs only the records in git, then the comparison with `results/`	every table and figure of this note except those needing `fetch-data`
`smoke`	engine validation: a tiny 1D campaign with bitwise resume; exact enumeration and bitwise restart in $d = 2, 3$	that the samplers are exact on small systems
`eval-fixed`, `eval-p1`	the 2D/3D evaluators under the frozen rule and under P1	tables 15, 17; appendix B
`calibration`	`calibrate_p1.py` (1 780 tasks, ≈ 7 min on 16 cores)	appendix C and the coverage numbers
`scan1d`, `tables1d`	`scan_v3_report.py`, `make_tables_and_figure.py`	tables 5, 6, appendix A
`refit1d DS=...`	`select_model_v3.py` on the released snapshot of one coupling (6–156 s on 20 cores); needs `fetch-data`	the fits, crossing series and label of that coupling; section 6.4.1
`note-tables`	`make_note_tables.py -check`	every generated line of this note
`fetch-data`, `build`, `note`	downloads the data not in git; the engines; this note (three passes)	–

12.3 Verified reproduction

On 7 October 2026 the folder, in its layout before the public release, was assembled afresh on the workstation and every target was run through Slurm. The engines built and all tests passed. The 2D/3D evaluations reproduced the pinned production evaluations in all per-point fields (maximum relative difference 3×10^{-11}), and so did the P1 evaluations and the calibration regenerated from scratch (2×10^{-11}). The 1D cross-check study, the 20-point 1D table and the 1D report tables were reproduced, and the 1D fitting stage reproduced 18 of 20 labels and all values (section 6.4.1). The comparison reports those two labels as known to be platform-sensitive.

On 13 October 2026 the folder in its public layout was exported afresh and tested through Slurm on two platforms: a personal desktop (Python 3.12.3, NumPy 1.26.4, SciPy 1.11.4) and an Expanse debug node (Python 3.8.8, NumPy 1.20.1, SciPy 1.6.2, GCC 10.2.0). On both, the engines build, all tests pass (3.1

and 7.5 min), and `make replay` (14 and 43 s) reports 9 checks PASS, none FAIL and one NOT RUN, the regenerated calibration, which needs `make calibration`; `note-tables` finds all 355 generated lines in this note. The cited 1D refits need `fetch-data` and were not part of this test.

12.4 What is not in git

`make fetch-data` downloads two archives from the project's Zenodo data record, doi:10.5281/zenodo.23466766, and checks their SHA-256 (`data/HOSTED.json`). `refit1d` needs the first: the 1D block archives of the selected snapshots (1.43 GB) and their merged summaries (25 MB). The second holds the per-degree fits of the selected runs (140 MB), for inspection; no target needs them. Regenerating the records from scratch needs the raw data: the 1D trajectories (0.655 TB) and the 2D/3D trees of chains, packed histograms and analyses (257.5 and 237.2 GiB). They are not hosted; they are available from the author on request. The records were produced from these by `analyze.py` and `merge_summaries.py` ($d = 1$) and by `pack_histograms.py` and `precision_analysis.py` ($d = 2, 3$).

A $d = 1$ frozen window parameters

The window parameters $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$ were frozen per coupling from earlier fits of the stage A/B data (section 5); h' is the N2 temperature half-width.

Table 20: Frozen $(\beta_c^{\mathrm{pre}}, \Delta_\varepsilon^{\mathrm{pre}})$ per coupling and N2 half-widths h' (units 10^{-4}) at $N = 2^{16}$ and 2^{19}.

Δ_σ	s	β_c^{pre}	$\Delta_\varepsilon^{\mathrm{pre}}$	$h'(2^{16})$	$h'(2^{19})$	N2
0.0125	0.975	0.596534	0.7910	14.67	9.50	no
0.025	0.95	0.558588	0.7179	10.83	6.02	no
0.0375	0.925	0.527150	0.6575	8.42	4.13	no
0.05	0.9	0.499660	0.6234	7.31	3.34	no
0.0625	0.875	0.474858	0.5898	6.35	2.71	yes
0.075	0.85	0.452099	0.5674	5.79	2.35	no
0.0875	0.825	0.430939	0.5462	5.30	2.06	no
0.1	0.8	0.411095	0.5292	4.94	1.86	yes
0.1125	0.775	0.392343	0.5191	4.74	1.74	yes
0.125	0.75	0.374529	0.5074	4.51	1.62	yes
0.1375	0.725	0.357528	0.4991	4.36	1.54	yes
0.15	0.7	0.341247	0.4930	4.25	1.48	yes
0.1625	0.675	0.325592	0.4863	4.13	1.42	yes
0.1727335	0.654533	0.313200	0.4845	4.10	1.40	yes
0.175	0.65	0.310504	0.4843	4.10	1.40	yes
0.1875	0.625	0.295925	0.4814	4.05	1.38	yes
0.2	0.6	0.281805	0.4808	4.04	1.37	yes
0.2125	0.575	0.268101	0.4775	3.98	1.34	yes
0.225	0.55	0.254785	0.4802	4.03	1.37	yes
0.2375	0.525	0.241813	0.4809	4.04	1.37	yes

B $d = 2, 3$ extrapolation series

For every coupling and series: Δ_ε of the extrapolation, ω' and its $\Delta\chi^2 \leq 1$ interval, χ^2/dof, σ with ω' fixed and profiled per draw (P1), eligibility.

Table 21: $d = 2$ series (P1 evaluation; central values identical to the frozen evaluation).

s	series	Δ_ε	ω' [interval]	χ^2/dof	σ fixed	σ profiled	eligible
1.05	U4	0.9750	0.56 [0.20, 2.00]	0.51/1	0.0020	0.0083	no: ω' edge
	U6	0.9719	0.84 [0.20, 2.00]	0.29/1	0.0014	0.0080	no: ω' edge
	U8	0.9700	0.96 [0.20, 2.00]	0.14/1	0.0011	0.0087	no: ω' edge
	Y	0.9208	0.60 [0.43, 0.76]	0.44/1	0.0018	0.0177	yes
	U4+Y	0.9840	0.26 [0.26, 0.27]	8.33/4	0.0037	0.0042	no: χ^2
1.1	U4	0.9670	1.17 [0.20, 2.00]	0.11/1	0.0011	0.0060	no: ω' edge
	U6	0.9663	1.33 [0.20, 2.00]	0.10/1	0.0010	0.0060	no: ω' edge
	U8	0.9651	1.41 [0.39, 2.00]	0.03/1	0.0009	0.0067	no: ω' edge
	Y	0.9285	0.80 [0.61, 1.00]	1.10/1	0.0015	0.0101	yes
	U4+Y	0.9720	0.35 [0.35, 0.36]	15.28/4	0.0028	0.0031	no: χ^2

$d = 2$ series, continued

s	series	Δ_ε	ω' [interval]	χ^2/dof	σ fixed	σ profiled	eligible
1.2	U4	0.9654	0.20 [0.20, 2.00]	3.57/1	0.0045	0.0043	no: χ^2
	U6	0.9657	0.20 [0.20, 1.44]	3.66/1	0.0044	0.0042	no: χ^2
	U8	0.9686	0.20 [0.20, 0.78]	2.73/1	0.0040	0.0044	no: χ^2
	Y	0.9806	0.34 [0.20, 0.60]	0.00/1	0.0027	0.0237	no: ω' edge
	U4+Y	0.9620	0.55 [0.53, 0.57]	5.85/4	0.0016	0.0018	yes
1.3	U4	0.9595	2.00 [0.20, 2.00]	0.49/1	0.0007	0.0030	no: ω' edge
	U6	0.9595	2.00 [0.20, 2.00]	0.56/1	0.0007	0.0029	no: ω' edge
	U8	0.9595	2.00 [0.20, 2.00]	0.70/1	0.0007	0.0036	no: ω' edge
	Y	0.9714	2.00 [1.58, 2.00]	0.72/1	0.0007	0.0012	no: ω' edge
	U4+Y	0.9832	0.20 [0.20, 0.20]	201.09/4	0.0042	0.0042	no: χ^2
1.4	U4	0.9624	0.20 [0.20, 2.00]	2.38/1	0.0042	0.0035	no: χ^2
	U6	0.9643	0.20 [0.20, 2.00]	2.30/1	0.0041	0.0038	no: χ^2
	U8	0.9689	0.20 [0.20, 1.32]	1.83/1	0.0039	0.0044	no: ω' edge
	Y	0.9868	0.20 [0.20, 0.76]	1.85/1	0.0042	0.0051	no: ω' edge
	U4+Y	0.9730	0.20 [0.20, 0.20]	93.97/4	0.0040	0.0040	no: χ^2
1.5	U4	0.9759	0.20 [0.20, 0.98]	0.48/1	0.0042	0.0055	no: ω' edge
	U6	0.9777	0.20 [0.20, 0.92]	0.46/1	0.0041	0.0056	no: ω' edge
	U8	0.9819	0.20 [0.20, 0.91]	0.29/1	0.0039	0.0064	no: ω' edge
	Y	0.9810	0.20 [0.20, 0.27]	3.15/1	0.0038	0.0055	no: χ^2
	U4+Y	0.9792	0.20 [0.20, 0.20]	18.48/4	0.0038	0.0038	no: χ^2
1.6	U4	0.9799	0.20 [0.20, 0.97]	1.09/1	0.0041	0.0053	no: ω' edge
	U6	0.9822	0.20 [0.20, 1.10]	0.59/1	0.0041	0.0057	no: ω' edge
	U8	0.9839	0.25 [0.20, 1.47]	0.18/1	0.0031	0.0063	no: ω' edge
	Y	0.9919	0.20 [0.20, 0.23]	6.34/1	0.0035	0.0050	no: χ^2
	U4+Y	0.9915	0.20 [0.20, 0.20]	56.29/4	0.0035	0.0035	no: χ^2
1.7	U4	0.9984	0.20 [0.20, 0.66]	0.51/1	0.0039	0.0087	no: ω' edge
	U6	1.0012	0.20 [0.20, 0.75]	0.50/1	0.0038	0.0093	no: ω' edge
	U8	0.9935	0.36 [0.20, 0.97]	0.57/1	0.0021	0.0097	no: ω' edge
	Y	1.0311	0.20 [0.20, 0.23]	5.82/1	0.0034	0.0038	no: χ^2
	U4+Y	1.0260	0.20 [0.20, 0.20]	225.45/4	0.0034	0.0034	no: χ^2

Table 22: $d = 3$ series (P1 evaluation).

s	series	Δ_ε	ω' [interval]	χ^2/dof	σ fixed	σ profiled	eligible
1.55	U4	1.4812	0.75 [0.44, 1.11]	0.00/1	0.0017	0.0163	yes
	U6	1.4809	0.79 [0.51, 1.10]	0.02/1	0.0016	0.0139	yes
	U8	1.4790	0.92 [0.71, 1.15]	0.02/1	0.0013	0.0075	yes
	Y	1.4486	1.05 [0.99, 1.11]	0.02/1	0.0014	0.0058	yes
	U4+Y	1.4789	0.78 [0.78, 0.79]	44.76/4	0.0015	0.0017	no: χ^2
1.6	U4	1.4726	1.29 [0.93, 1.74]	0.28/1	0.0011	0.0072	yes
	U6	1.4720	1.35 [1.02, 1.75]	0.29/1	0.0011	0.0062	yes
	U8	1.4709	1.39 [1.14, 1.67]	0.37/1	0.0010	0.0040	yes
	Y	1.4750	1.21 [1.14, 1.28]	1.51/1	0.0013	0.0048	yes
	U4+Y	1.4731	1.24 [1.23, 1.25]	2.38/4	0.0011	0.0012	yes
1.7	U4	1.4737	0.66 [0.28, 1.09]	0.07/1	0.0017	0.0168	yes
	U6	1.4724	0.74 [0.40, 1.14]	0.03/1	0.0015	0.0157	yes
	U8	1.4700	0.92 [0.67, 1.20]	0.02/1	0.0012	0.0082	yes
	Y	1.5414	1.16 [1.08, 1.24]	0.02/1	0.0012	0.0050	yes
	U4+Y	1.4862	2.00 [2.00, 2.00]	7052.05/4	0.0008	0.0008	no: χ^2
1.75	U4	1.4587	1.69 [1.03, 2.00]	2.89/1	0.0009	0.0062	no: χ^2
	U6	1.4583	1.75 [1.17, 2.00]	3.47/1	0.0009	0.0047	no: χ^2
	U8	1.4581	1.65 [1.28, 2.00]	3.84/1	0.0009	0.0033	no: χ^2
	Y	1.5654	1.35 [1.27, 1.44]	4.12/1	0.0011	0.0041	no: χ^2
	U4+Y	1.6546	0.20 [0.20, 0.20]	10550.66/4	0.0050	0.0050	no: χ^2
1.8	U4	1.4553	2.00 [1.53, 2.00]	2.02/1	0.0009	0.0016	no: χ^2
	U6	1.4551	2.00 [1.59, 2.00]	2.37/1	0.0009	0.0015	no: χ^2
	U8	1.4542	2.00 [1.62, 2.00]	2.76/1	0.0008	0.0016	no: χ^2
	Y	1.5918	1.55 [1.46, 1.64]	5.46/1	0.0010	0.0030	no: χ^2
	U4+Y	1.6897	0.20 [0.20, 0.20]	11986.04/4	0.0050	0.0050	no: χ^2
1.85	U4	1.4583	1.20 [0.71, 1.85]	0.04/1	0.0010	0.0075	yes
	U6	1.4589	1.17 [0.76, 1.68]	0.01/1	0.0010	0.0062	yes
	U8	1.4592	1.19 [0.92, 1.50]	0.00/1	0.0009	0.0045	yes
	Y	1.6306	1.39 [1.30, 1.48]	0.20/1	0.0010	0.0034	yes
	U4+Y	1.6685	0.20 [0.20, 0.20]	14600.83/4	0.0045	0.0045	no: χ^2
1.9	U4	1.4545	1.15 [0.63, 1.87]	0.89/1	0.0011	0.0072	yes
	U6	1.4544	1.23 [0.78, 1.81]	0.83/1	0.0011	0.0057	yes
	U8	1.4546	1.29 [1.00, 1.64]	0.83/1	0.0009	0.0034	yes
	Y	1.6563	1.63 [1.53, 1.74]	2.91/1	0.0009	0.0026	no: χ^2
	U4+Y	1.6896	0.20 [0.20, 0.20]	13764.62/4	0.0046	0.0046	no: χ^2
1.95	U4	1.4500	2.00 [1.56, 2.00]	0.49/1	0.0008	0.0014	no: ω' edge
	U6	1.4498	2.00 [1.61, 2.00]	0.39/1	0.0008	0.0013	no: ω' edge
	U8	1.4490	2.00 [1.64, 2.00]	0.05/1	0.0007	0.0014	no: ω' edge
	Y	1.6827	2.00 [1.88, 2.00]	0.70/1	0.0008	0.0014	no: ω' edge
	U4+Y	1.6808	0.20 [0.20, 0.20]	19130.26/4	0.0048	0.0048	no: χ^2

C P1 calibration strata

Each cell: eligible synthetic fits: profiled $1\sigma/2\sigma$ coverage (fixed-ω' 1σ coverage), status (P pass, F fail, nc not calibrated, i.e. fewer than 200 eligible fits). Last column: synthetic sets with U_4 and Y both eligible: false-failure rate of the profiled cross-check (fixed), status.

D Protocol identities and history

SHA-256 prefixes of the files whose identity matters (full hashes in `protocols/` and in the records): FIT_MODEL_V3 rev. 14 `f05b835b`; FIT_MODEL_2D rev. 5 `4d2efc63`; FIT_MODEL_3D rev. 5 (approved) `b4ccf436` and rev. 7 (production) `682858f0`; amendment P1 rev. 4 `2a18c0ae`; frozen evaluators `evaluate_2d.py` rev. 5 `0fe221e8` and `evaluate_3d.py` rev. 1 `6bd35d7a`; P1 evaluators rev. 6 `3c3e35fc` and rev. 2 `d635c917`; `calibrate_p1.py` `d94b59f2`; calibration `CALIBRATION.json` `897142c1`; production evaluations `fbfc0ef9` (2D) and `ae9cf237` (3D); P1 evaluations `f9272978` (2D) and `b3fd6364` (3D). Records that named a personal machine were scrubbed after the results and re-pinned; `RE-PINNED.json` lists the original and published hashes (for example `CALIBRATION.json`: original `897142c1`, published `1afae79a`, which differ only in a host label). The prefixes above are the original ones.

The file `FIT_MODEL_3D.md.sha256` records the hash at approval (revision 5), while the evaluators pin revision 7. The approved revision 5, revision 6 and the difference to revision 7 are released (`protocols/FIT_MODEL_3D_rev5_approved.md`, ...); the two differ only in the preflight condition and the gate records described in section 5. `FIT_MODEL_2D.md` has no approval hash because it was never approved. The status lines inside the protocol files (for example "UNREVIEWED") predate their reviews; `PROTOCOL_HISTORY.md` gives the review status of each revision.

E The pre-campaign ε-expansion reference

Before the campaign, six-loop reference values were computed and frozen with the records (`data/1d/reference/published_and_epsilon_reference.csv`); they played no role in any selection. "GFF only" (generalized free field, i.e. the Gaussian end) is the ordinary Borel–Padé [4/2] of the six-loop series about the Gaussian end; "endpoint assisted" is the Borel–Padé [3/3] of the remainder R in $\Delta_\varepsilon = 1 - \sqrt{1-e} + (1-e)/8 + (1-e)^{3/2} R(e)$, $e = \varepsilon$, which builds in the $s \to 1$ endpoint behavior. They used an older value of the six-loop coefficient of the $d = 1$ series, $C_6(1) = 14331.8 \pm 0.5$; the final value, 14331.687, is given in the accompanying article [1]. They are superseded by the eight-loop values of section 6.5. The same file contains columns from an exploratory pre-campaign analysis (`new_MC_conditional_fit` and related), which are also superseded.

F Glossary

Δ_σ, Δ_ε dimensions of the spin and of the leading $\mathbb{Z}_2$-even scalar; $1/\nu = d - \Delta_\varepsilon$.

Δ_{eff}^{Q} finite-size effective dimension from a doubling pair at the U_4 crossing, eq. (4).

σ_{stat}, σ_{sys}, σ_D bootstrap error of the quoted value; half-range over the model family (a sensitivity scale, not a standard deviation); error of a paired difference.

p, k_τ polynomial degree of the scaling function; order of the thermal field ($d = 1$ fits).

P_{χ^2} χ^2 probability of a fit.

coupling one value of s (or Δ_σ) of the simulation grid; inverse temperatures are β.

window, variant, configuration in $d = 1$: the range $|x| \le x_{\max}$ of the scaling variable kept in a fit; the form of the correction terms (`leading`, `second`, `background`, `log`); one choice of window, variant, $N_{\min}$ and k_τ (section 6.2).

gate a pass/fail condition on a fit or series.

quotable, eligible a $d = 1$ fit that passes its gates; a $d = 2, 3$ extrapolation series that passes its gates.

series $\Delta_{\text{eff}}^{Q}(L)$ over the doubling pairs for one Q ($d = 2, 3$) or one crossing window and degree rule ($d = 1$), extrapolated to $L \to \infty$.

U4+Y joint extrapolation of U_4 and Y with common Δ_ε and ω'.

x-check cross-check (table 2).

Table 23: Calibration strata of amendment P1 (top $d = 2$, bottom $d = 3$).

$d = 2$, s	U_4	U_6	U_8	Y	U4+Y	cross-check
1.05	0 nc	0 nc	19: 1.00/1.00 (0.16) nc	1493: 0.83/0.95 (0.12) P	1812: 0.68/0.95 (0.59) P	0 nc
1.1	0 nc	1: 1.00/1.00 (0.00) nc	8: 1.00/1.00 (0.00) nc	1672: 0.83/0.95 (0.15) P	1799: 0.69/0.96 (0.63) P	0 nc
1.2	0 nc	0 nc	0 nc	490: 0.33/0.82 (0.00) F	1805: 0.67/0.95 (0.62) P	0 nc
1.3	0 nc	0 nc	0 nc	196: 1.00/1.00 (0.01) nc	194: 0.10/0.44 (0.01) nc	0 nc
1.4	0 nc	0 nc	0 nc	0 nc	15: 0.87/1.00 (0.87) nc	0 nc
1.5	1: 0.00/1.00 (0.00) nc	0 nc	3: 0.00/1.00 (0.00) nc	234: 0.00/0.02 (0.00) F	0 nc	0 nc
1.6	0 nc	0 nc	2: 1.00/1.00 (0.00) nc	222: 0.00/0.03 (0.00) F	0 nc	0 nc
1.7	62: 0.00/0.13 (0.00) nc	69: 0.00/0.10 (0.00) nc	144: 0.40/1.00 (0.00) nc	214: 0.00/0.04 (0.00) F	0 nc	0 nc

$d = 3$, s	U_4	U_6	U_8	Y	U4+Y	cross-check
1.55	1225: 0.86/0.98 (0.19) P	1396: 0.87/0.97 (0.17) P	1666: 0.83/0.96 (0.17) P	1663: 0.69/0.95 (0.20) P	1771: 0.68/0.95 (0.63) P	1062: 0.007 (0.69) P
1.6	1122: 0.99/1.00 (0.31) P	1165: 0.98/1.00 (0.31) P	1378: 0.88/1.00 (0.29) P	1668: 0.69/0.95 (0.21) P	1786: 0.70/0.95 (0.67) P	1076: 0.007 (0.61) P
1.7	825: 0.92/1.00 (0.20) P	1107: 0.90/1.00 (0.19) P	1603: 0.89/0.97 (0.16) P	1671: 0.69/0.96 (0.18) P	218: 0.29/0.89 (0.06) F	108: 0.000 (0.43) nc
1.75	428: 1.00/1.00 (0.01) P	420: 0.98/1.00 (0.02) P	684: 0.81/1.00 (0.15) P	1710: 0.69/0.95 (0.23) P	47: 0.00/0.09 (0.00) nc	108: 0.000 (0.76) nc
1.8	228: 0.99/1.00 (0.00) P	246: 0.95/1.00 (0.01) P	229: 0.41/1.00 (0.02) F	1645: 0.70/0.96 (0.26) P	1: 0.00/0.00 (0.00) nc	79: 0.000 (0.87) nc
1.85	717: 1.00/1.00 (0.32) P	996: 1.00/1.00 (0.27) P	1483: 0.93/1.00 (0.23) P	1695: 0.70/0.96 (0.22) P	0 nc	93: 0.108 (1.00) nc
1.9	531: 1.00/1.00 (0.38) P	764: 1.00/1.00 (0.32) P	1289: 0.98/1.00 (0.32) P	1663: 0.71/0.97 (0.29) P	0 nc	92: 0.326 (1.00) nc
1.95	216: 1.00/1.00 (0.00) P	237: 0.99/1.00 (0.01) P	254: 0.47/1.00 (0.00) F	182: 0.15/0.84 (0.01) nc	0 nc	39: 0.641 (1.00) nc

Table 24: $d = 1$: the frozen six-loop reference values against the Monte Carlo result.

Δ_σ	$\varepsilon = 1 - 4\Delta_\sigma$	six-loop, GFF only	six-loop, endpoint assisted	MC (this work)
0.025	0.900	0.5703	0.7109	0.71115(49)(959)
0.05	0.800	0.5459	0.6182	0.61987(55)(220)
0.0625	0.750	0.5351	0.5871	0.59003(35)(150)
0.1	0.600	0.5085	0.5262	0.53248(20)(242)
0.15	0.400	0.4871	0.4897	0.49325(41)(306)
0.1727335	0.309	0.4829	0.4836	0.48368(52)(78)
0.2	0.200	0.4828	0.4829	largest pair

$[\sigma]$ label suffix: $\sigma(\Delta_{\text{eff}}^{U_4})$ of the largest pair above the target 0.003.

frozen fixed before the data in question were analyzed (a protocol, the $d = 1$ window parameters); "the frozen rule" is the $d = 2, 3$ protocol before amendment P1.

fixed-ω' / profiled rule bootstrap at the central ω' (the frozen rule) / ω' re-profiled in every draw (amendment P1).

preflight short runs at the production sizes, before a production, that check rates, autocorrelation times and crossing brackets.

stage 0, A, B; N0, N2, N3 1D campaigns (table 4); N3, an extension to $N = 2^{20}$, was not run.

N1 the first evaluation of all 20 $d = 1$ couplings with protocol v3 (section 5).

snapshot an immutable, hashed set of merged summaries from which one fit run is made.

CB8 the eight-loop ε expansion of the accompanying article, resummed by conformal Borel resummation with the short-range end imposed.

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